林口長庚醫院兒童血液腫瘤科病房 **Orientation**

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2016.9 Revised

<兒童血液科Orientation> (Ed. 2016.9)

Admission note:

勿使用取出暫存或ditto，**請用前一次主治醫師修改過的出院病摘的present illness**

Measurable goals 本科範本：Admission for chemotherapy, neutropenic fever, Leukemia workup

JCI 評鑑要求：Data請有判讀, ROM/PE請與主訴/PI相符, Imp不可用縮寫,主訴有疼痛請加上疼痛評估

Progression note

勿使用取出暫存，請參考主治醫師修改版本(有⊕記號的, not T)，Subjective, Assessment每天不同，請至少有主治醫師迴診意見

Summary要condense，尤其是長期住院病人，切忌每週或每次交接班都是Copy前次病歷

Special Procedure：如有幫忙完成IT or BMA, 請完成，範本在"special procedure" -> "本科範本"->"TIT order" "BMA order"

出院照護計劃書：請點"範本"，並**註明回診時間及醫師**

出院摘要：**Diagnosis不可用縮寫**，出院指示請帶入出院照護計劃，請整理檢驗及檢查格式，勿**只是帶入**

同意書

本科常用同意書：骨髓檢查，兒童腰椎穿刺檢查或治療，中重度鎮靜，輸血治療（每次住院第一次），化學治療（新診斷病人），造血幹細胞移植同意書

電腦斷層，磁振照影，chloral hydrate，**請盡量以電子簽章完成**

醫藥囑：除當日住院病人外，**星期五所有病人請"Order renew" (不只Medication)**

常用本科醫囑範本：

1. Admission order: for入院化療病人使用
   病房vital signs routine: **TPR QID, "I/O Q8H", BP QD 請勿更動**！
   大部份化療病人均會在門診抽好CBC/DC，確認血球指數可進行化療才會住院
   住院後需檢驗Cr, ALT及U/A作為化療前基本Survey，並請**確認是否真的有三天內的CBC/DC data**
2. Neutropenic fever：True EMERGENCY
   Vital signs建議BP Q8H，以提早偵測septic shock，病房routine就是I/O Q8H
   IVF建議以至少half saline以上的濃度來維持vessel tonicity: 組套設定為D2.5/0.45S
   Fever routine: "拜託請勿出現塞屁股的退燒藥"，以免造成肛門黏膜損傷進一步造成perianal abscess
   另外住院身體檢查也請多注意屁股口腔mucosa及全身皮膚容易潛藏感染處
   Timely Antibiotics is important. Neutropenic fever病人一旦住院，請放下任何手邊工作
   第一時間把抗生素藥囑開立完成親手交給護理師送單，當做emergency處理，
3. BMA order 及 TIT order （含藥囑範本）
   請幫忙完成同意書，白血病做BM study請附剩餘檢體同意書兩張（均須收回）
   並注意鎮靜藥物劑量，請以mg表示劑量，**請勿以PC or mL表示 (Ketamine 1pc = "500"mg!)**
4. Discharge order:
   除非主治醫師特別註明，每位病人出院均開立CBC/DC單子回門診追蹤血球指數，請註明回診日期抽血
   當次住院移植之病人，回診要加驗生化Cr & K (因使用Cyclosporine緣故)
   住院打high dose MTX的case，因為會影響藥物排除，在住院間會hold 口服Bactrim使用
   若病人平常有在使用，出院時請記得開回給病人帶藥回家服用 BID QW1,3,5 for PJP prophylaxis
   有些病人會預先開立預約住院單，請註明於出院醫囑中，並從HIS住院醫囑中"常用功能""開立住院通知單"中
   點選"預住院日期""主診斷""重大傷病身份""14日內再住院"，再存檔列印
5. 住院範本TPOG-ALL-2013 HD-MTX SR or HR/VHR
   ALL入院進行鞏固期Consolidation治療，每個病人需進行每兩周一次，四次共八週的治療
   每次住院需進行髓鞘內化學治療(TIT: triple intrathechal chemotherapy，俗稱打背)
   睡前口服6MP，以及高劑量Methotrexate 24小時infusion (2.5g/m2 for SR, 5g/m2 for HR/VHR)
   MTX輸注從晚上十點開始，劑量的1/10一小時內給完，其餘9/10在23小時內滴完
   一入院就會hydration + alkalization with jusomin
   MTX開始滴注後42小時(第三天下午四點)，會檢驗血漿MTX濃度，並開始使用leucovorin解毒every 6 hours
   若濃度超過一定值，也會調整leucovorin的劑量，之後每24小時檢驗濃度，直至濃度< 0.1umol/L便可回家
   出院帶藥包括6MP +/- Bactrim，並安排住院日起算兩周後的門診及住院
   第一次鞏固治療，須同時安排打背及骨髓穿刺檢查，以評估引導期的療效，確認是否已緩解(remission)
   開立住院醫囑時，請點選admission order**包含藥物及化療醫囑範本**，第二次以後的chemo order可用ditto

*病房常備IVF: D0.225S (500) (1000), D2.5/0.45S (500), N/S (500)，其餘均需從U/D藥局，也就是藥囑開立
每個病房IVF keep open定義不同，請盡量不使用keep open, 一律25ml/hr minimal for port-A line

*備輸血：本單位只使用LPR, LPP, ILPR, ILPP，不使用PRBC or platelet concentration (開刀除外)

輸血請註明血品 用量 日期 輸注多久。

例：LPR 2U on 12/1 run > 2 hours 或 LPP 6U on 12/1 and 6U on 12/2, run > 1 hour

每次住院的第一次輸血，請開立輸血同意書。

異體移植病人，移植前中後期，或SAA under immunosuppressive therapy，請使用照光血品ILPR or ILPP

移植後病人：備血單上請於**"特殊備血需求"**註明**"預輸血型，血品，donor血型及recipient血型"**

ILPP及ILPR建議分開開備血單，以減少血庫ILPP備血所需時間

有頻繁輸血出疹的病人，需開立premedication包括Diphenhydramine +/- hydrocortisone。但前次過敏不代表這次一定會過敏

*重大傷病開立：三種表單給書記：乙種診斷證明書（需有正確重大傷病ICD-10代碼）、重大傷病申請單、骨髓檢查或病理報告

ICD10代碼：X = 看部位

ALL C9100, AML (M0, M1, M2, M7) C9200, AML (M4) C9250, AML (M5) C9300

Brain tumor C71X; Retinoblastoma: C692X

Lymphoma: Lymphoblastic lymphoma C835X, DLBCL C833X, Burkitt C837X, ALCL ALK+ C846X

Hodgkin lymphoma: C81XX (看組織型及位置)

Neuroblastoma: adrenal C741X, retroperitoneum C480; Kidney (Wilms): C64X

Germ cell tumor (ovary): C56X; Hepatoblastoma: C222;

Bone (Osteosarcoma, Ewing): C40XX Soft tissue (RMS or …): C49XX; SAA D619

*末期病人：安寧共照：楊金美護理師＆高振益醫師。每週填寫關懷評估表、timing of DNR。Expire：死診、乙診、遺體處理單

*PortA注意事項：

Heparin 目前有三種劑型：100u/cc total 10cc and 10u/cc total 3cc 給拔portA使用的heparin lock

另外一種1pc 25000u的原汁是拿來IVF用的,在6L病房目前只有BMA會用到(rinse針筒用)

PortA拔針時，若非現拔現on，須以heparin沖管預防管路阻塞，但使用過量仍可能產生heparin induced thrombocytopenia

>40kg: 100u/ml 5cc push; 10~40kg: 100u/ml 3cc push; <10kg: 10u/ml 3cc push

請參考本科醫囑範本： “01Heparin Lock “ 開立 （含藥物）

*****PortA 阻塞處理*****：先確定是否有走針，若走針拔掉重on；

若有把握在正確位置上，Urokinase 6000U N/S diluted to 3cc PortA lock for at least 2 hours

*藥囑開立常見陷阱！！ **病房常備藥：Diphenhydramine, labetalol, lasix, valium, luminal, codeine, morphine, ketamine**

針劑Diphenhydramine點選時，預設為”IM”，請改為IV；Codeine 預設為IM, 但在病房使用大多為IVF or IV

升血球針Rhg-csf有兩種劑型：300 mcg通常for 40kg以上，30kg 以下請開立75 mcg劑型 (5mcg/kg)

另外點選時，預設為”SC”，請改為IV，以減少針孔 (但是Donor的g-csf就通常為SC，效果較持續)

兒癌基金會補助藥(不吃健保)：Zofran 8mg/tab, “4mg”/amp, Kytril 1mg/tab, “1mg”/amp, 口服A酸，藥囑打「兒癌」就可以找到

Fentanyl貼片有三種劑量：12.5 mcg/hr/patch, 25mcg/hr/patch, 50mcg/hr/patch (0.5~1mcg/kg/hr)

病房常備的止痛藥為Codeine及morphine，若考慮其他種止痛，請審慎考慮領藥時間等因素

QOD & Q3D 藥物請註明「給藥日期」

*高貴藥物 (健保價/自費價)

*IVIG: for transplant (0.5g/kg自費,Day -7, +7, +21, +35, +56, +77, +98), HLH 1g/kg x 2 days or ITP 1g/kg x 1-2 day, 3g/PC from 血液基金會 NT $5400/$6480

*Posaconazole: for adolescent, 5cc tid for prophylaxis, 10cc bid for treatment, 105cc/PC NT $12811/$14092; 100mg/tab NT $696/$766

*Voriconazole (Vfend) 200mg/vial NT $4430/$4873; 200mg/tab NT $1266/$1393; 50mg NT $361/$397

*Caspofungin (Cancidas) 50mg/vial NT $8367/$9204

*Liposomal amphotericin B (AmBisome) 50mg/vial NT $6172/$6800

*Ganciclovir 500mg/vial $1510/$1661 *Valganciclovir 450mg/tab &840/$969

*Cytomegalovirus antibody (Cytotect) 50ml/bot (100U/ml) NT $31387/$34526

*Cyclosporine syrup: 100mg/cc, 50cc/pc NT $7941/$8735

*Irinotecan 100mg/vial 自費 NT $6384; *Topotecan 4mg/vial 自費 NT $9962;

*Dactinomycin 0.5mg/vial 自費 NT $4704; *Mitoxantrone 20mg/vial 自費 NT $4576

*Busulfan 60mg/10ml/amp NT $7314/$8390; *ATG 25mg/vial NT $7631/$8394;

*Fludarabine 50mg/vial 自費 NT $6978; *Clofarabine 20mg/vial 自費 NT $55000;

*Bortezomib 3.5mg/vial 自費 NT $4211 *Temodal 100mg/cap NT $3655/$4021, 20mg/cap NT $769/$846

*G-CSF 300mcg/pc NT $2040/$2329; 75mcg/pc NT $763/$862.4;

*Eltrombopag 25mg/tab 自費 NT $1478

*Rituximab 500mg/50ml/vial NT $38887/$43176, 100mg/10ml/vial NT $8132/$9029

*Imatinib (Glivec) 100mg/tab NT $626/$694.9 ; Dasatinib (Sprycel) 20mg/tab NT $543/$622; 50mg/tab NT $1354/$1501

*化療醫囑注意事項

除藥單外請記得開立處置單（從小兒科檢查進入）並標注批價日期，IV一天批一張 （<1hr, 1~4hr, 4~8hr, >8hr）

化療藥局白班上班便開始準備調配當日10:30以前送至藥局的藥單，由助理員11:30前後領回

若臨時因病況需DC or hold chemotherapy, 請速電洽2915請藥師停止配藥，若已經調配完來不及阻止，便只能棄藥或轉寄賣

若來的及DC，請再由化療醫囑開立DC chemotherapy並註明退藥量以利書記退賬

當天預進行的chemotherapy請盡可能在上午完成開立，不然小夜才能領藥而小夜人力又不足，易造成危險

IVF的Cyclosporine及ganciclovir 也由化學藥局調配，配合作業時間及領藥，

cyclosporine若Q12H給藥時間為18:00, 06:00, Q8H為13:00, 21:00, 05:00

若遇到調整劑量或預計改為口服，請注意給藥時間及退藥問題

*TPN醫囑注意事項

TPN品項繁多，在本病房Neonate PN, Peripheral PN, Pediatric standard, Standard, Hepatic都可能用到

開立前請與主治醫師確認清楚品項及劑量

處方送單時間為11:00前截止，11:00前開立的藥物當日下午三點左右便可回病房

超過此時間開的TPN都視為隔天處方，若有急需請開立STAT (並聯絡TPN藥局2704確認是否有存量)

*檢驗醫囑注意事項

PB smear: 小兒科檢查 -> Blood smear interpretation (M32-504) **(沒有兒童血液科字樣)**

移植前需配對的個案，recipient with potential sibling donor：開立HLA typing, 開立代碼為 L72-654, L72-659

若要找unrelated donor需高解析解碼：開立High resolution HLA-A,B,C,DR genotyping L72-666 **自費NT 12,000**

Chimerism test (STR: short tandem repeat) L72-869: 移植前recipient & donor, 移植後植入時&視情況追蹤分析cell來源比例，可選Blood or BM

移植中病人需監測CMV: 請驗以下之一

優先：CMV DNA quantitative amplification test (L72-966)，病房固定週一抽血，二五操作，二下午可知

"Blood" Cytomegalovirus PP65 Ag (L72-922)，病房固定週一抽血，下午四點多可知data

與驗Serology IgG/IgM, Rapid CMV culture (urine)都不同

移植後病人若懷疑PTLD，或HLH病人: 請驗EBV DNA定量測試 L72-998，每週三上午操作，四點多可知data

Blood fungus culture 請勿開"Fungus culture"，請開立 Blood culture 把檢體別改成"黴菌血液培養"

*移植病人 care

異體移植: Donor 來源: Matched sibling donor (MSD) PBPCT, MUD PBPCT (unrelated), and Cord blood transplantation (CBT)

住院後移植前評估：dental condition, echocardiogram & EKG, CXR, viral serology data before transfusion, PICC line or Hickman line, Chimerism test

調理治療 Conditioning therapy: Chemotherapy (Busulfan, cyclophosphamide) +/- Total body irradiation (TBI) and anti-thymoglobulin (ATG)

移植當日：Stem cells transfusion (total nucleated cell dose? CD34 dose?), premedication, DMSO dose if cryopreserved stem cell (移植病房無菌室在 11F), Donor's STR

GvHD prophylaxis: cyclosporin, methotrexate, steroid; Balance between GvHD and GvL (graft versus leukemia)

Neutrophils engraftment: First day of three days with ANC > 500; 10~20 days in PBPCT, longer for CBT, STR analysis

Platelet engraftment: First day of >20K without further transfusion

Post-transplant care: in positive-pressured protective isolation ward, Hematopoietic support, Monitor GvHD, Opportunistic infection prophylaxis: bacteria, fungus and virus, BP control, nutrition support, medication compliance

（大小孩的口服 acyclovir 800mg 對面長青買一罐 3300 NT，開立請用藥囑中「自備藥口服」後面備註 Acyclovir 800mg）

Serious complication: severe GvHD, invasive fungal disease, viral reactivation, PTLD, sinusoidal obstruction syndrome, graft failure, autoimmune cytopenia

林口長庚醫院兒童血液腫瘤科病房 Orientation

Hematological manifestation of systemic diseases

1. Blood diseases: diagnosis and treatment of anemia, bleeding disorders, and WBC disorders
2. Oncologic diseases: tumors, hepatosplenomegaly, lymphadenopathy
3. Hematopoietic stem cell transplantation

Characteristics of our patients:

1. Differential diagnosis
2. Seriousness of underlying diseases
3. Chronic nature of disease
4. Emergency nature
5. Painful procedure: bone marrow aspiration and biopsy, lumbar puncture and intrathecal chemotherapy, implantation of central venous catheter (Port-A catheter), delivery of chemotherapeutic agents
6. Toxicities of anticancer drugs
7. Use of treatment protocols
8. Psychosocial issues: include physician's role in communicating with patients and parents, different stages of patients (new patient, critical patient, dying patient)
9. Cancer staging, and cancer registry
10. Tumor board, multidisciplinary team care
11. Supportive care
12. Continuous care (in-patient and out-patient services)

Patient assessment:

1. Good history taking and physical examination
2. Chart review, accurate medical records including admission note, progress note, discharge summary, and data sheet
3. Frequent hand washing

References:

1. Textbook of Pediatrics (Nelson's), 20th edition, 2015
2. Nathan and Oski's Hematology and Oncology of Infancy and Childhood, 2-Volume Set, 8th edition, 2014
3. Principle and Practice of Pediatric Oncology (Pizzo and Poplack), 7th edition, 2015
4. Manual of Pediatric Hematology/Oncology (Lanzkowsky), 6th edition, 2016
5. Thomas' Hematopoietic Cell Transplantation, 5rd edition, 2016
6. RED BOOK, American Academy of Pediatrics
7. Physician's Desk Reference (PDR)
8. 輸血醫學 (孫建峰醫師著)
9. http://www.cancer.gov/publications/pdq/information-summaries/pediatric-treatment PDQ® Cancer Information Summaries: Pediatric Treatment 美國國衛院參考資料

Frequently used telephone numbers or cell phone numbers (常用分機與 GSM 號碼)

6L 護理站	8519, 8520
BMT room	2302, 2303
Cell marker (血清免疫)	8365 轉 365
Cytogenetics (chromosome)	2355 (Ms 青燕)
Blood bank	2538, 2548
HLA typing, STR (分生組)	8360 轉 360
MRI	8401
6L 血液抹片室	8195 (Ms 素秋)
6L 化療室	8196
6F 化療藥局	2915
施麗雲醫師研究室	2364
癌症分子診斷實驗室(造血癌症組)	5672, 5673
洪悠紀醫師	0965275103
楊兆平醫師	辦公室 8216, GSM 65912
江東和醫師	辦公室 8226, GSM 65911
黃一安醫師	辦公室 8968, GSM 65954
陳世翔醫師	辦公室 8966, GSM 65913
張從彥醫師	GSM 68263
專科護理師　余婷娟護理師	GSM 68292
兒癌個管師　楊淑賀護理師	GSM 67742
移植協調師　溫玉娟護理師	GSM 67658
血液科　施麗雲醫師	GSM 60107
小兒外科　賴勁堯醫師	GSM 65969
小兒外科　明永青醫師	GSM 65970
小兒外科　楊曉雯醫師	GSM 66183
小兒神經外科　吳杰才醫師	GSM 65520
放射診斷科　王超然醫師	GSM 65760
放射診斷科　呂嘉偉醫師	GSM 65742
放射診斷科　黃敏政醫師	GSM 60776
放射診斷科　張志丞醫師	GSM 66378
病理科　薛純醫師	GSM 65829
病理科　容世明醫師	GSM 66020
放射腫瘤科　曾振淦醫師	GSM 65779
放射腫瘤科　詹頂立醫師	GSM 66934
放射腫瘤科　高偉恆醫師	GSM 68294
骨科　施信農醫師	GSM 65586
眼科　高玲玉醫師	GSM 65863

Taiwan Pediatric Oncology Group (TPOG) protocols

Acute lymphoblastic leukemia (ALL)	TPOG ALL 2013
	TPOG ALL 2002 SRA
	TPOG ALL 2002 SRB
	TPOG ALL 2002 HR
	TPOG ALL 2002 VHR
	TPOG ALL 2002 VHR for infant
	Relapse protocol
Acute myeloid leukemia (AML)	TPOG AML 97A
	TPOG AML 2008
	TPOG APL 2001
Non-Hodgkin's lymphoma	TPOG NHL 2010
Neuroblastoma	TPOG N2002
Wilms' tumor	TPOG W97 & W2016
Osteosarcoma	TPOG OS2k
Rhabdomyosarcoma	TPOG RMS 2007
Langerhan's cell histiocytosis	TPOG-LCH2003
Hemophagocytic lymphohistiocytosis	TPOG-HS2003

http://www.ccfroc.org.tw/child/index_child.php

中華民國
兒童癌症基金會



Routine activities in Pediatric Hematology/Oncology:

1. Daily chart rounds and ward round
2. 每週一上午 7:30-8:30 Morning meeting (at 6L 討論室)
3. 每月第一週週三上午 11:00-12:00 Orientation (at 6L 討論室)
4. 每月第二週週三上午 11:00-12:00 Case conference (at 6L 討論室)
5. 每月第四週週三上午 11:00-12:00 Journal meeting (at 6L 討論室)
6. 每月第三週週四下午 3:00-4:00 Tumor conference (at 醫學大樓 3 樓病理科討論室)
7. Reading of blood smears (at 6L 血液抹片室)
8. Scheduled teaching activities by attending physicians

Procedures performed at 6L 血液抹片室

1. PB smear for morphology (M32-504)
2. Bone marrow aspirate interpretation (M32-501, M32-503) and cytochemical stains (M32-512, M32-513, M32-517, M32-518)
3. Bone marrow biopsy (M32-502)
4. Lymph node or bone marrow biopsy imprint interpretation (M32-505)
5. Reticulocyte count (M32-536)
6. CSF or other body fluid cytology (cytospin) (M32-519)
7. Osmotic fragility test (M32-532)
8. Betke stain (staining for HbF in blood smear) (M32-522)
9. BM iron stain (M32-511)
10. LAP score (M32-515)
11. Sugar Water test (M32-534)
12. Acid Ham's test (M32-535)
13. Bleeding time (IVY method) (M32-563)
14. PT (M32-568)
15. aPTT (M32-569)
16. PTT substitution test (M32-572)
17. Thrombin time (M32-567)
18. Protamine gel test (3P test) (M32-583)
19. Fibrinogen quantitative (M32-566)
20. Thrombo-Wellcotest (FDP) (M32-584)
21. D-dimer (M32-090-Z)
22. DIC profile (M32-586)
23. Heinz body preparation (M32-537)
24. Hb H preparation (M32-531)
25. Microhematocrit (M32-540)

Taking a pediatric history

--- The history is the single most important factor in arriving at a correct diagnosis

1. Source of history and reason for referral
2. Identifying information
3. **Chief complaint (CC)**
4. History of present illness
5. **Birth history**: antenatal, natal, neonatal
6. **Development**: milestone, urinary continence, control of defecation, comparison of development with that of siblings and parents, any period of failure to grow or unusual growth, schoold grade, quality of work
7. **Nutrition**: breast or formula feeding, supplements, solid food, appetite
8. Past medical history
9. **Immunizations and tests**: type, number, reactions
10. Operations
11. Accidents and injuries
12. Medications
13. Family history
14. **Personal history**: relation with other children, school progress
15. **Social history**: family structure, family support systems, child care arrangement, school, insurance
16. **Habits**: sleeping, recreation, elimination, behavioral concern, adolescent habit, dental hygiene, safety, family health habit as model
17. **Review of systems** (ROS): general review, skin, eyes, ENT, dental, cardiopulmonary system, gastrointestinal system, genitourinary system, neuromuscular system, endocrine system, hematopoietic system

兒童血液腫瘤科注意事項

1.新發病病童:
開立重大傷病証明: 住院醫囑系統→常用功能→開立証明書→重大傷病申請開立(住院醫師開立蓋章即可)
2.新發病病童癌症治療計劃書登入:
請主治醫師登入住院醫囑系統→治療中心→癌症中心
→癌症治療計畫書第一頁: 基本資料完成(含診斷、本院組織細胞病理診斷、STAGING SYSTEM OR CLINICAL TNM、預後因子、共病、癌症既往史及其他) →完成後下一頁/儲存
→癌症治療計畫書第二頁 :治療計畫及討論要點→ENTER 後→TREATMENT 畫面(點化療 PROTOCOL 查詢) →請點 PEDIATRIC EPIRUBICIN BASED REGIMEN 並註明療程→存檔(新發病打 EPIRUBICIN)
一定要登錄才能開藥

化療醫囑注意事項
1.普通 HYDRATION 3000ML/M2/DAY
2.若 PROTOCOL 有指定 HYDRATION 用法依 PROTOCOL 建議開醫囑
3.VINCRISTIN MAX 2MG(勿超過 2MG)
4.高劑量 ARA-C：請配合 DEXAN + ACETAMINPHEN + FLUOROMETHOLONE (眼藥水)
5.高劑量 CYCLOPHOSPHAMIDE AND IFOSFAMIDE：請配合 HYDRATION + MESNA
6.高劑量 MTX：請配合 LEUCOVORIN ,安全 MTX LEVEL < 0.1 , LEUCOVORIN 才能停
7.打 ASPARAGINASE：PLEASE CHECK URUNE SUGAR IN EVERY OTHER VOIDING URINE
8.打 CISPLATIN：請配合 MANNITOL + KCL + MGSO4 + CAL.GLUCONATE
9.大部分化療劑量以體表面積計算：若 BAS>1.73M2 (以 1.73M2 算)；若體重< 10 公斤，請示主治醫師是否須改成公斤來計算 （原體表面積劑量 ÷ 30 = 每公斤劑量）
10.有些化療劑量特別註明公斤來計算，請注意!
11.TIT 劑量以年齡來算而非體表面積來計算
13.高劑量 MTX IVF 請開 1Gm/40ML/VIAL
14.第一次 TIT 送檢須含 CSF ROUTINE、CSF CYTOLOGY、CSF 生化(PROTEIN、SUGAR、LACTATE)
15.VINCRISTIN AND EPIRUBICIN COUNTINE INFUSION 給予須用 PORT-A LINE
16.VP-16 稀釋液請用 0.9%生理食鹽水 OR D5W (不會沉澱)

6L 白血病病人抽取骨髓檢體注意事項:

Medication:

1. Sedation
   - ●Midazolam 0.1mg/kg/dose (若病人可以配合，可以不需要)
   - ●Ketamine 1mg/kg/dose (若病人可以配合，可以不需要)
2. 局部麻醉藥: Lidocaine 1% 5ml/amp　1PC
3. Heparin 25,000U/5ml/vial 1PC

Equipment:

1. 骨髓穿刺針、骨髓穿刺包 (若用導尿包，請再多準備刀片)
2. 5mL 空針筒(for 局部麻醉藥)，23 號針頭 2 支，20ml 空針筒至少 3 支
3. 無菌試管 x1，綠頭管 x3，紫頭管(大)x2
4. 隔離衣 x1，無菌手套 x1
5. O2 standby

Document:

1. 請家屬簽署骨髓檢查同意書 x1 (請用住院醫囑系統列印)
2. 請家屬簽署剩餘檢體同意書 x2 (病房表單櫃內有)
3. 癌症分子診斷單 x 1
4. MRD measurement sheet x 1

新診斷或復發　約 18ml

1. 先抽 heparin 來 rinse 20ml 空針 2 支
2. 穿刺針到達骨髓後，先抽少許做 smear，確定有 particle 後再接續以下部分
3. 無菌試管 2ml for leukemia marker
4. 綠頭管 2ml for chromosome
5. 紫頭管(大) 2ml for molecular study

（3–5）可一次抽 6ml, 之後再各分配 2ml

6. 綠頭管 5 ml For 陳醫師 study
7. 綠頭管 5 ml For MRD study
8. 紫頭管(大) 2ml For MRD study

若 dry tap，而 PB blasts > 70%，改抽 PB 綠頭 10mlx2 管+紫頭管(大) 2ml (可再詢問陳醫師)

追蹤　約 9ml

1. 先抽 heparin 來 rinse 20ml 空針 1 支
2. 穿刺針到達骨髓後，先抽少許做 smear，確定有 particle 後再接續以下部分
3. 紫頭管(大) 2ml x 2 管
4. 綠頭管 5 ml

若有任何疑問, 請 call 陳世翔醫師 GSM65913.

新診斷白血病注意事項：Approaching a patient with newly diagnosed leukemia

1. 詳細的病史詢問+理學檢查
2. 可點選範本 Leukemia work-up (Admission order ± Daily order)
3. 點滴給 D5S1/4 500cc + 20cc NaHCO3 run 3000ml/m2/day (**若 P 離子濃度高，請馬上移除含有 NaHCO3 的點滴，因鹼性環境下易形成磷酸鈣結晶！**)
4. Routine 檢驗項目:
   - CBC/DC, PT/aPTT, Fibrinogen, anti-thrombin III
   - BUN, Cr, AST, ALT, Na, K, Ca, P, uric acid, LDH, albumin, total protein
   - EBV-VCA IgG, CMV IgG, HBsAg, anti-HBs Ab, anti-HBc Ab
   - IgG, IgM, IgA, U/A
   - BLOOD SMEAR INTERPRETATION
   - CXR
   - 備血
5. 骨髓檢查項目:
   - BONE MARROW ASPIRATION
   - BONE MARROW INTERPRETATION
   - B.M. PEROXIDASE STAIN & INTERPRETATION
   - B. M. P.A.S. STAIN & INTERPRETATION
   - SPECIFIC & NON-SPECIFIC STAIN (診斷 AML 時需要)
   - ACUTE LEUKEMIA MARKER (送 7J 血清組)
   - CHROMOSOME STUDY (送成人血液科)
   - RT-PCR (送成人血液科)：ALL 請開 **C77-001** 共 3 張，AML 請開 **C77-029** 共 1 張
   - MRD study: **C77-043**
6. Medications:
   - Allopurinol # po tid (10mg/kg/day)
   - Rasburicase 0.15-0.2mg/kg/dose, max. 7.5mg (uric acid＞8)(**禁忌症： G6PD deficiency**)
   - Antibiotics if indicated
7. 若病人的 WBC＞100000，請小心輸血，維持血小板＞20000，若無症狀，盡量不輸紅血球，因為會增加血液黏滯性；若有嚴重貧血心衰竭症狀，可輸少許紅血球，且放慢輸血速度。

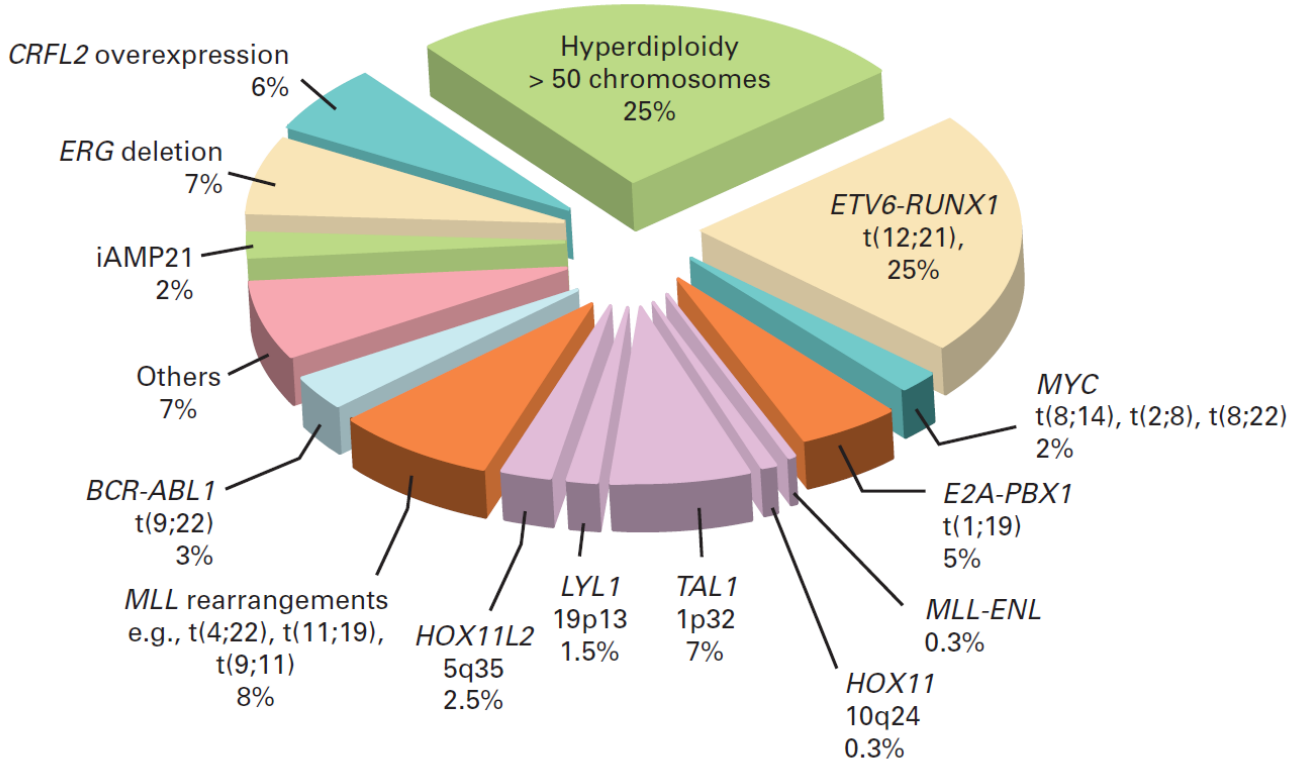
Characteristics and clinical outcomes of selected subtypes of childhood ALL

Subtype	Frequency (%)	Clinical Implication	Estimated 5-yr EFS (%)
B-cell precursor			
Hyperdiploidy > 50	20–30	Excellent prognosis with antimetabolite-based therapy	85–95
t(12;21)(p13;q22) ETV6-RUNX1	15–25	Expression of myeloid-associated antigens CD13 and CD33; excellent prognosis with intensive asparaginase therapy	80–95
Trisomies 4 and 10	20–25	Excellent prognosis with antimetabolite therapy	85–90
t(1;19)(q23;p13) TCF3-PBX1	2–6	Increased incidence in blacks; excellent prognosis with high-dose methotrexate treatment; increased risk of CNS relapse in some studies	80–85
Intrachromosomal amplification of chromosome 21	2–3	More common in older children and adolescents; poor prognosis; benefit from intensive induction and early re-intensification therapy	30–40
t(4;11)(q21;q23) MLL-AF4	1–2	Poor prognosis and predominance in infancy, especially those < 6 months of age; overexpression of FLT3	30–40
t(9;22)(q34;q11.2) BCR-ABL1	2–4	Imatinib plus intensive chemotherapy improve early treatment outcome	80-90 at 3 years
t(8;14)(q23;q32.3)	2	Favorable prognosis with short-term intensive therapy with high-dose methotrexate, cytarabine, and cyclophosphamide	75–85
Hypodiploidy < 44 chromosomes	1–2	Poor prognosis	35–40
CRLF2 overexpression	6–7	Poor prognosis; common in patients with Down syndrome (55%)	?

Subtype	Frequency (%)	Clinical Implication	Estimated 5-yr EFS (%)
T-cell			
TAL/LMO rearrangement	15–30	Good prognosis in some studies; potentially responsive to histone deacetylase inhibitor	?
HOX11 rearrangement	7–8	Good prognosis	?
HOX11L2 (TLX3) rearrangement	20–24	Poor prognosis in some studies	?
OXA rearrangement	4–5	Poor prognosis; potentially sensitive to histone H3K79 methyltransferases inhibitor	?
NUP214-ABL1	6	Sensitive to tyrosine kinase inhibitor	~50 (survival)
MLL-ENL	2–3	Favorable prognosis	80–90
Early T-cell precursor	12	Poor prognosis; expressed myeloid or stem-cell markers	30–35
Cooperation mutations			
B-cell precursor			
IKZF1 deletions/mutations	15–30	Poor prognosis; resistant to asparaginase and daunorubicin	50–55
JAK mutations	?2–5	Predominance in high-risk patients; JAK2 mutations in 20% of Down syndrome cases; potentially responsive to JAK2 inhibitors	~60
T-cell			
NOTCH/FBXW7 mutations	50	Favorable prognosis; potentially responsive to NOTCH inhibitor	90
PTEN-P13K-AKT pathway	~50	?Poor prognosis	?
CDKN2A/2B deletions	~70	?Potentially responsive to DNA methyltransferases inhibitor	?

Pui CH et al. JCO 2011;29:551-565

Estimated frequency of specific genotypes in childhood ALL



Pui CH et al. JCO 2011;29:551-565

TPOG ALL 2013 RISK CLASSIFICATION

Patients are classified into one of three categories (standard-, high-, or very high-risk) based on the presenting age, leukocyte count, presence or absence of CNS-3 status or testicular leukemia, immunophenotype, cytogenetics and molecular genetics, DNA index, and early response to therapy. Hence, definitive risk assignment will be made after completion of remission induction therapy. The criteria and the estimated proportion of patients in each category are provided below.

3.CRITERIA FOR STANDARD RISK ALL

1.B-lymphoblastic ALL with DNA index ≥ 1.16 [or hyperdiploidy (51-68)], *TEL-AML*1 fusion, or age 1 to 9.9 years and presenting WBC < 50,000/mm^3. AND

2. Must not have:

-CNS 3 status (≥ 5 WBC/μL of cerebrospinal fluid with morphologically identifiable blasts or cranial nerve palsy).

-Overt testicular leukemia (evidenced by ultrasonogram).

-Adverse genetic features: t(9;22) or *BCR-ABL1* fusion; t(1;19) with *E2A-PBX1* fusion; rearranged *MLL* (as measured by FISH and/or PCR)*;* or hypodiploidy (< 44 chromosomes).

-Poor early response (≥ 1% lymphoblasts on day 15 of remission induction, ≥ 0.01% lymphoblasts by immunologic or molecular methods on remission date).

CRITERIA FOR HIGH-RISK ALL

1. All cases of T-cell ALL and those of B- lymphoblastic ALL that do not meet the criteria for standard-risk or very high-risk ALL.

CRITERIA FOR VERY HIGH-RISK ALL

1. t(9;22) or *BCR-ABL1* fusion (with MRD ≥0.01% after remission induction including dasatinib (60 mg/m^2 per day).

2. Infants with t(4;11) or *MLL* fusion.

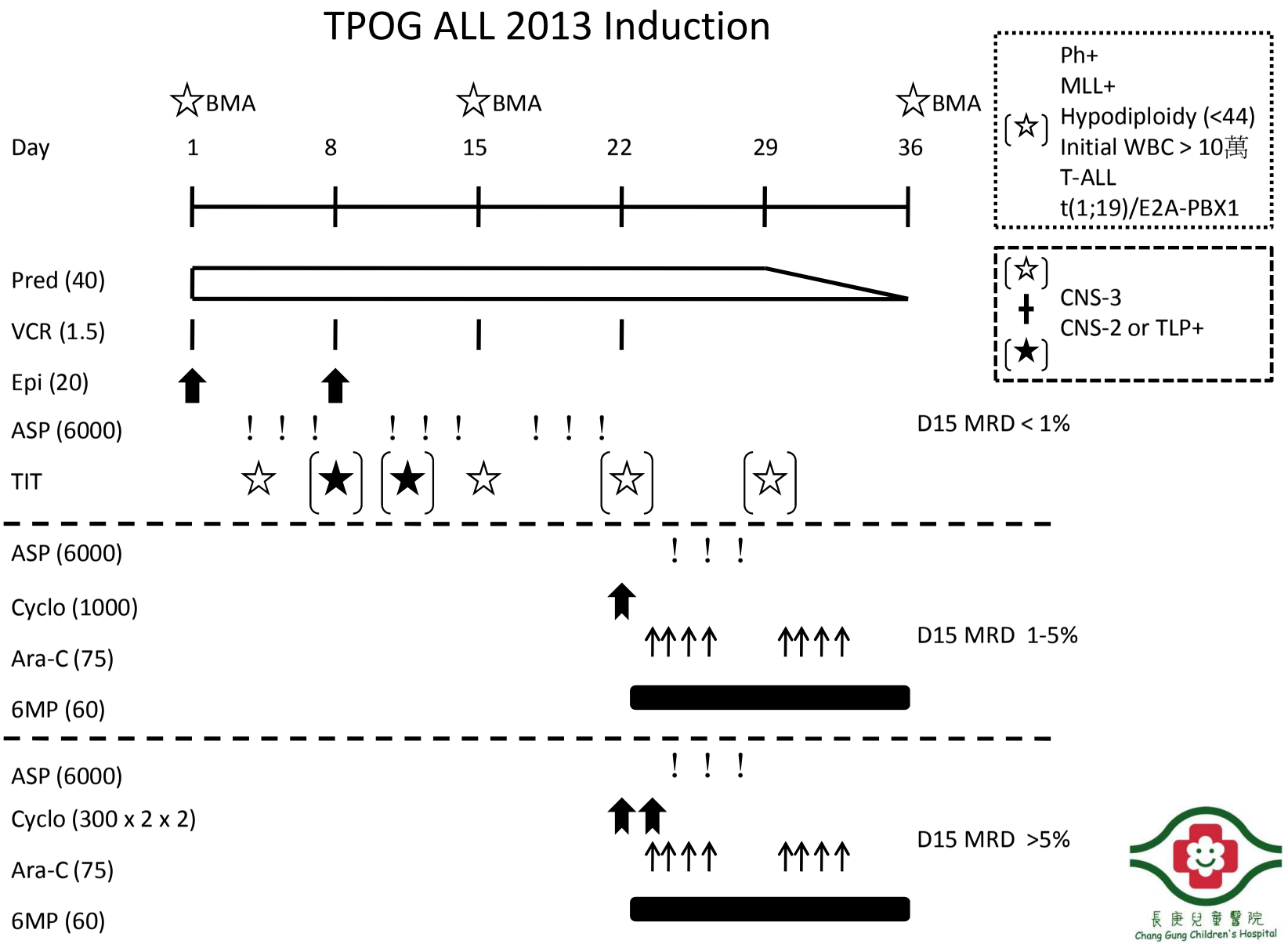
3. Induction failure or > 1% leukemic lymphoblasts in the bone marrow on remission date (with the exception of hyperdiploid (51-68) and *TEL-AML1* cases who should have positive MRD after consolidation therapy).

4. > 0.1% leukemic lymphoblasts in the bone marrow in week 7 of continuation treatment (i.e. before reinduction I, ~14 weeks post remission induction).

5. Re-emergence of leukemic lymphoblasts by MRD (at any level) in patients previously MRD negative.

6. Persistently detectable MRD at lower levels.

7.Early T-cell precursor (ETP) ALL, defined by lack of expression of CD1a and CD8 and low or absent expression of CD5 together with aberrant expression of myeloid and hematopoietic stem cell markers (such as CD13, CD33, CD34 and CD117).

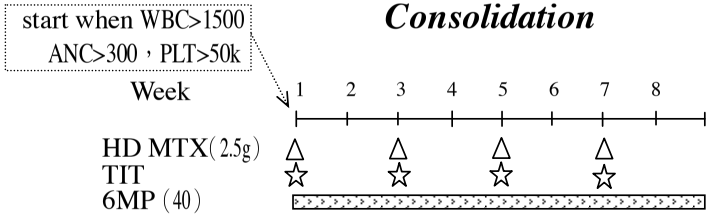

TPOG ALL 2013 Induction
☆BMA
☆BMA
☆BMA
Day
1
8
15
22
29
36
Pred (40)
VCR (1.5)
Epi (20)
ASP (6000)
TIT
ASP (6000)
Cyclo (1000)
Ara-C (75)
6MP (60)
ASP (6000)
Cyclo (300 x 2 x 2)
Ara-C (75)
6MP (60)
(☆) Ph+
MLL+
Hypodiploidy (<44)
Initial WBC > 10萬
T-ALL
t(1;19)/E2A-PBX1
(☆) + (★) CNS-3
CNS-2 or TLP+
D15 MRD < 1%
D15 MRD 1-5%
D15 MRD >5%
長庚兒童醫院
Chang Gung Children's Hospital

TPOG 2013 SR

Induction

start when WBC>1500
ANC>300，PLT>50k

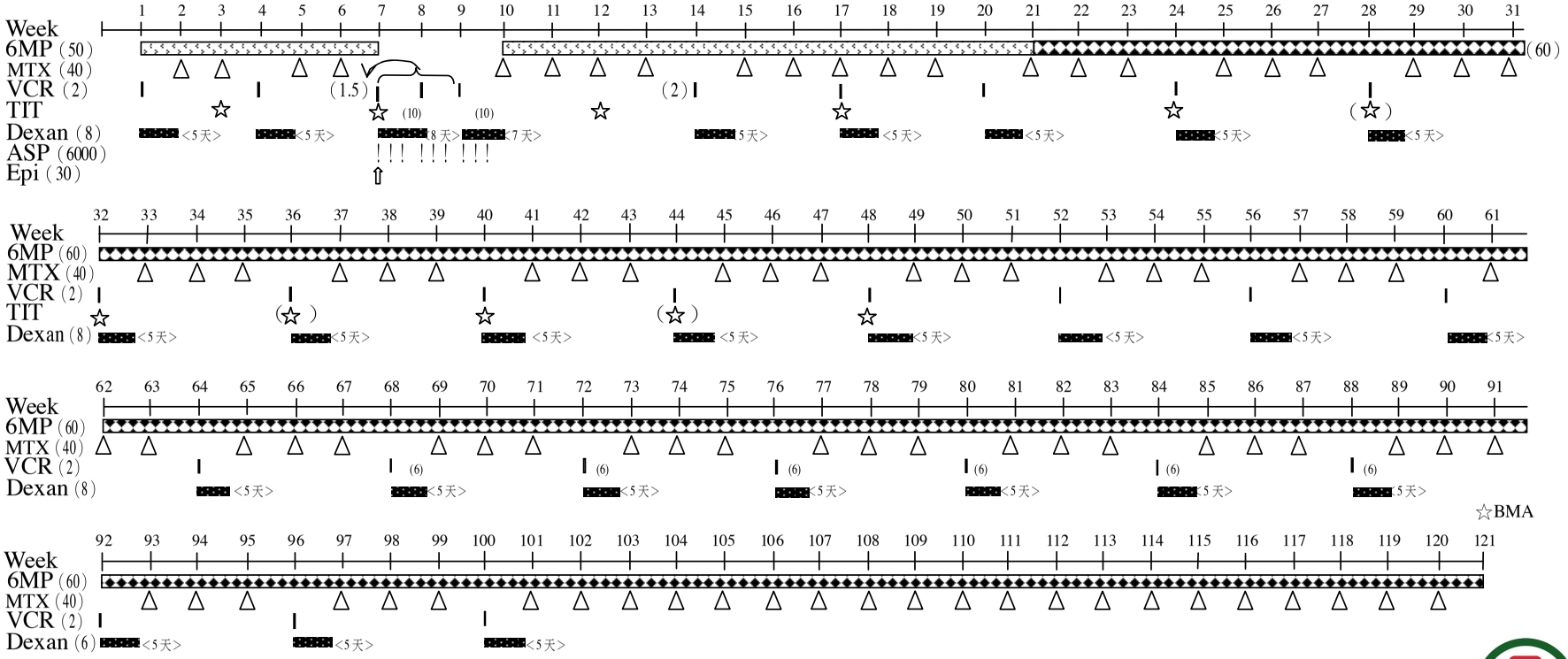
Consolidation



Rest 1 ~2 weeks till
ANC ≧ 300
WBC ≧ 1500
PLT ≧ 50k
No mocositis

姓名：
病歷號：

Continuation



長庚兒童醫院
Chang Gung Children's Hospital

Survival Analysis (from TPOG 2011 annual meeting)

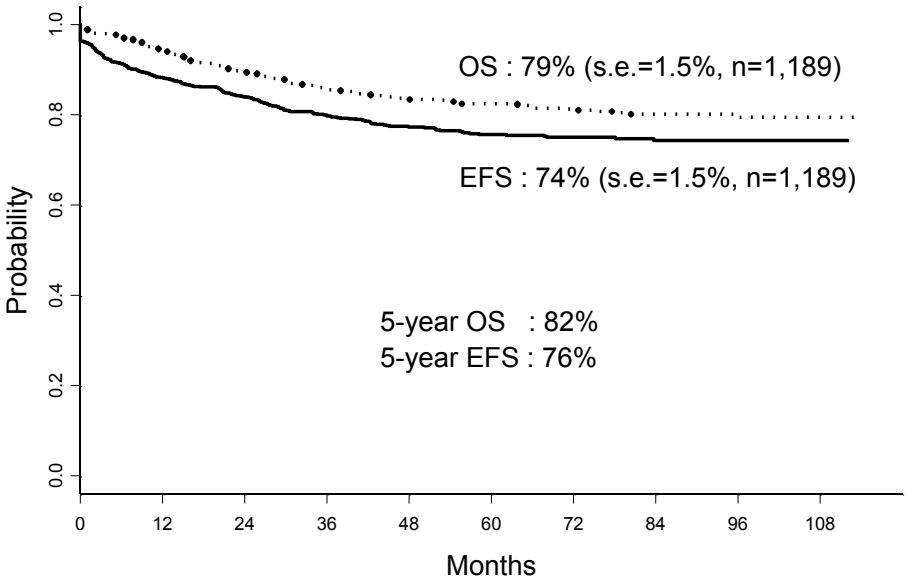


Fig 2. TPOG ALL2002. Overall Survival and event-free survivals. (2002.1~2011.8)

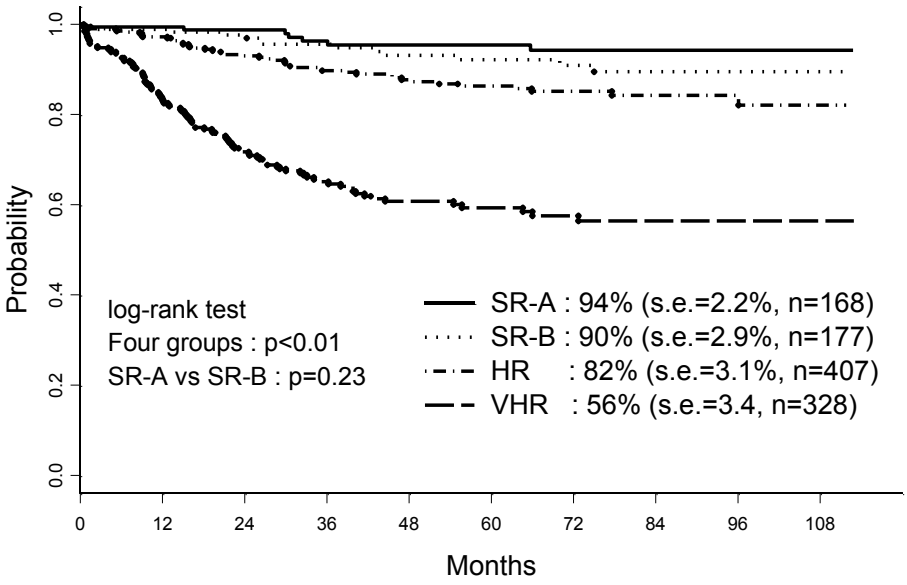


Fig 3. TPOG ALL2002. Overall survival by protocols. (2002.1~2011.8)

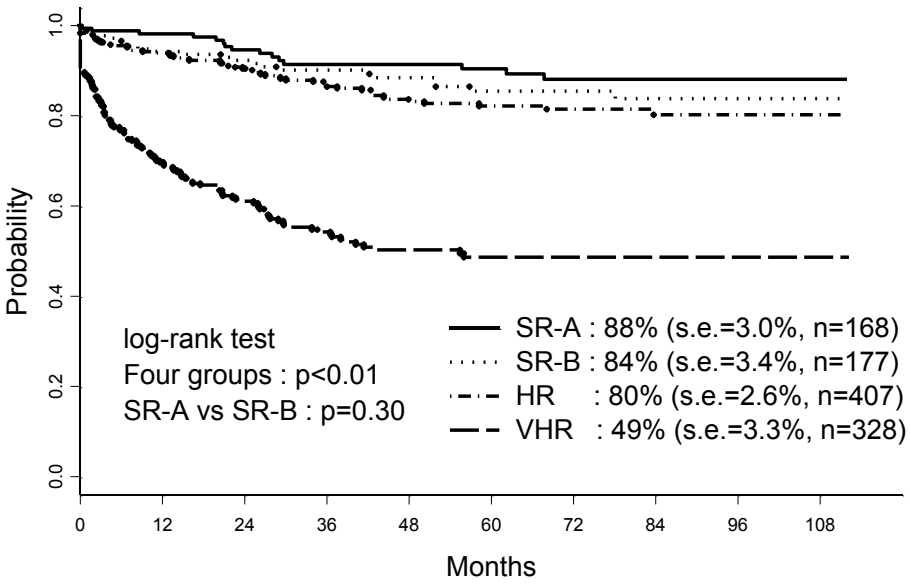


Fig 4. TPOG ALL2002. Event-free survival by protocols. (2002.1~2011.8)

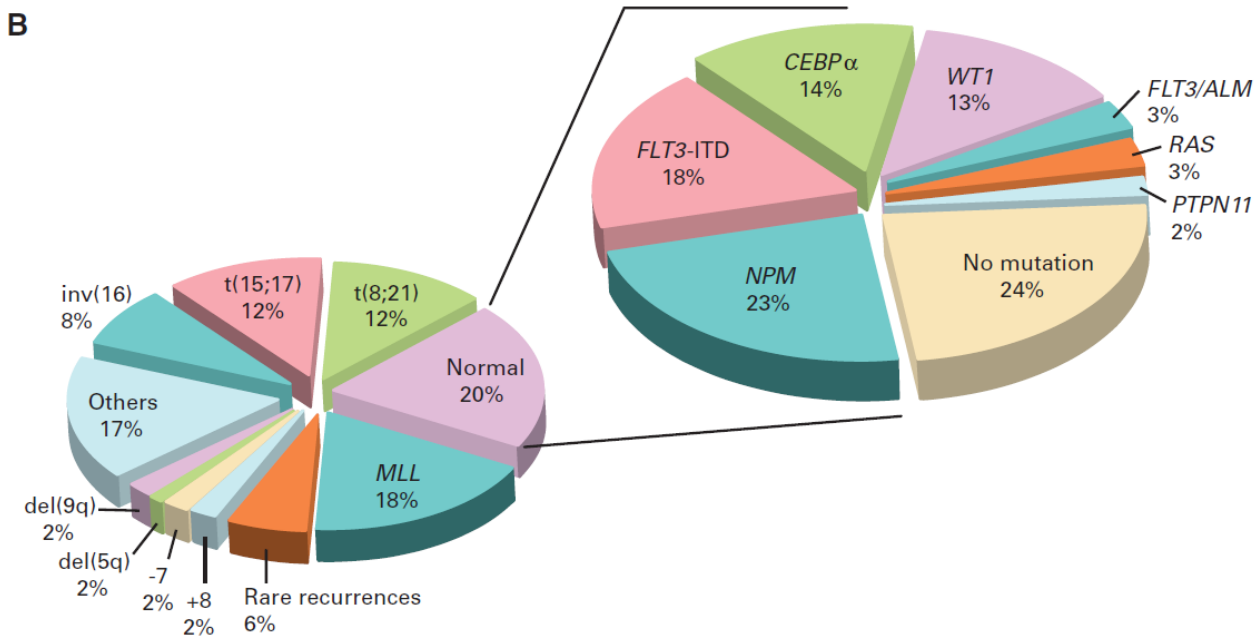
Characteristics and clinical outcomes of selected subtypes of childhood AML

Subtype	Affected gene	Frequency (%)	Clinical characteristics	Estimated 5-y Event-free Survival(%)
Rearrangement				
t(8;21)(q22;q22)	ETO-AML1	12	Associated with chloromas	55-71
inv(16)(p13;q22)	MYH11-CBF	8	Eosinophilia with dysplastic basophilic granules	72-88
t(8;16)	MOZ-CBP	1	High WBCs, chloromas, etoposide related secondary AML	ID
t(15;17)(q22;q12)	PML-RARA	12	Associated with FAB M3, Auer rods common; FAB M3; ATRA-sensitive	71
t(11;17)(q23;q12)	PLZF-RARA	rare	Associated with FAB M3, Auer rods common; FAB M3; ATRA-resistant	ID
t(1;22)	RBM15-MKL1	2-3	Associated with FAB M7 in Down syndrome and non–Down syndrome	ID
t(6;9)(p23;q34)	DEK-NUP214(CAN)	rare	Basophilia and multilineage dysplasia; associated with FLT3-ITD and TdT+	ID
MLL	MLL (partner genes)	18		
t(1;11)(q21;q23)	AF1q (MLLT11)	3	76% < age 2 years; 20% and 48% FAB M4 and M5, respectively	92
t(4;11)(q21;q23)	AF4 (MLLT2)	2	61% < age 2 years; 17% and 42% FAB M4 and M5, respectively	29
t(6;11)(q27;q23)	AF6 (MLLT4)	5	9% < age 2 years; 57% ≥ 10 years; 35% and 41% FAB M4 and M5, respectively	11
t(9;11)(p22;q23)	AF9 (MLLT3)	43	42% < age 2 years; 81% is FAB M5	50
t(10;11)(p11.2;q23)	AF10 (MLLT10)	2	75% < age 2 years; 27% and 55% FAB M4 and M5, respectively	17
t(10;11)(p12;q23)	AF10 (MLLT10)	13	62% < age 2 years; 72% is FAB M5	31
t(11;19)(q23;p13)	ELL or ENL (MLLT1)	4	58% < age 2 years; 42% and 45% FAB M4 and M5, respectively	49
t(11;19)(q23;p13.1)	ELL or ENL (MLLT1)	4	41% < age 2 years; 30% and 33% FAB M4 and M5 respectively	46
t(11;19)(q23;p13.3)	ELL or ENL (MLLT1)	3	36% < age 2 years; 44% ≥ age 10 years; 20% and 40% FAB M4 and M5	46
t(11;17)(q23;q21)	AF17 (MLLT6), LASP1	2	33% < age 2 years; 42% ≥ age 10 years; 33% and 50% FAB M4 and M5	11
Other		19	50% < age 2 years; 29% and 50% FAB M4 and M5, respectively	39

Subtype	Affected gene	Frequency (%)	Clinical characteristics	Estimated 5-y Event-free Survival(%)
Normal karyotype				
Gene mutations				
NPM	Nucleophosmin	23	8%-10% of childhood AML	65-80
CEBPα	CCAAT/enhancer binding protein α	14	4%-6% of all childhood AML; more common in older patients, FAB M1 or M2	70
FLT3/ALM	Fms-like tyrosine kinase 3 activation loop domain	3	6%-7% of all childhood AML	50-60
FLT3-ITD	Fms-like tyrosine kinase 3 internal tandem duplication	18	10%-15% of all childhood AML	< 35
WT1	Wilms tumor 1	13	8%-10% of all childhood AML	22-35
RAS	Rat sarcoma gene	3	5% of all childhood AML	ID
PTPN11	Protein tyrosine phosphatase, nonreceptor type 11	2	Most commonly associated with JMML	ID
No known mutations		24		40
Poor-risk cytogenetics		<15		<40
Del 5q/E5		1		
-7		2		

Pui CH et al. JCO 2011;29:551-565

Estimated frequency of specific genotypes in childhood AML



Pui CH et al. JCO 2011;29:551-565

FAB and WHO classification of acute myeloid leukemia

FAB classification

M0	AML with minimal differentiation
M1	Myeloblastic leukemia without maturation
M2	Myeloblastic leukemia with maturation
M3	Acute promyelocytic leukemia
M4	Acute myelomonocytic leukemia
M5	Acute monoblastic leukemia
M6	Acute erythroblastic leukemia
M7	Acute megakaryoblastic leukemia

WHO classification

AML with recurrent cytogenetic translocations
- AML with t(8;21)(q22;q22), AML1(CBF-α)/ETO
- Acute promyelocytic leukemia
- AML with t(15;17)(q22;q12) and variants; *PML/RAR*α
- AML with abnormal bone marrow eosinophils: inv(16)(p13;q22) or t(16;16)(p13;q22); CBF-β/MYH1
- AML with 11q23 (*MLL* gene) abnormalities

AML with multilineage dysplasia
- With prior MDS
- Without prior MDS

AML with myelodysplastic syndrome, therapy related
- Alkylating agent related
- Epipodophyllotoxin related
- Other types

AML not otherwise categorized
- AML minimally differentiated
- AML without maturation
- AML with maturation
- Acute myelomonocytic leukemia
- Acute monocytic leukemia
- Acute erythroid leukemia
- Acute megakaryocytic leukemia
- Acute basophilic leukemia
- Acute panmyelosis with myelofibrosis

Taiwan Pediatric Oncology Group (TPOG) protocols for AML

TPOG AML 97A

TPOG AML 97B (已結束)

TPOG AML 2008

TPOG AML or MDS in Down Syndrome

TPOG APL 97 (已結束) and TPOG APL 2001

Survival Analysis (from TPOG 2011 annual meeting)

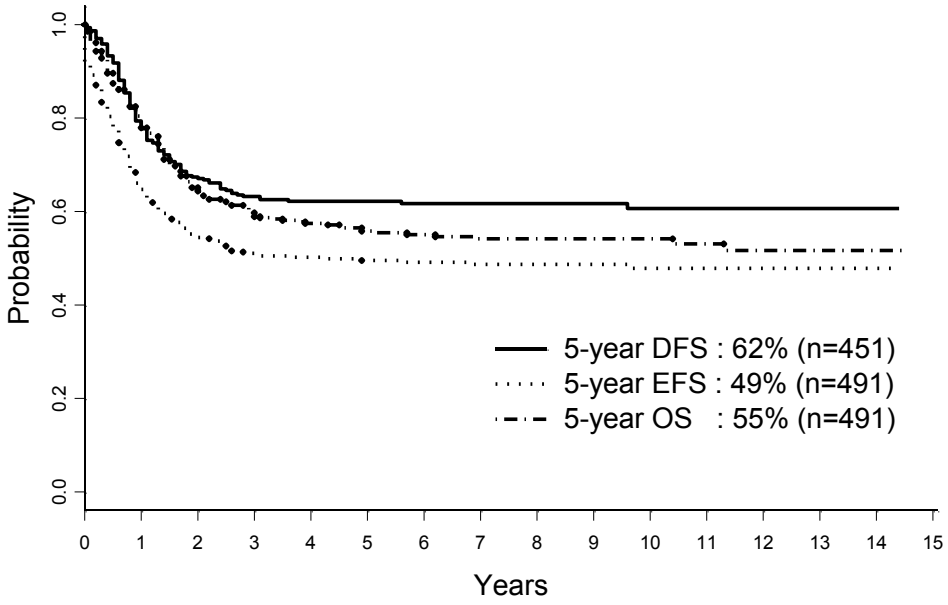


Fig 4. TPOG AML97. Overall, event-free and disease-free survivals. (1997.1~2011.8)

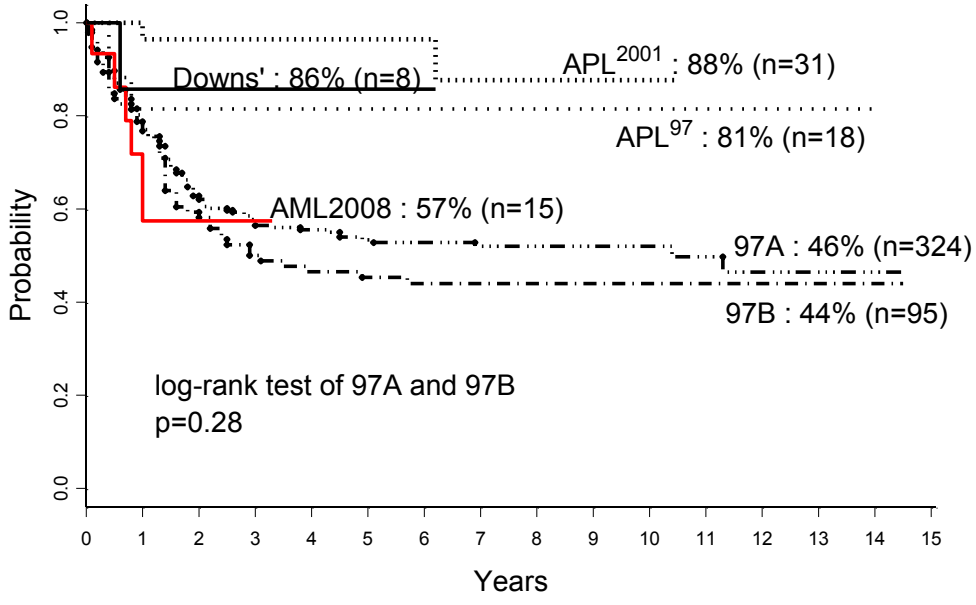


Fig 5. TPOG AML97. Overall survival by protocol. (1997.1~2011.8)

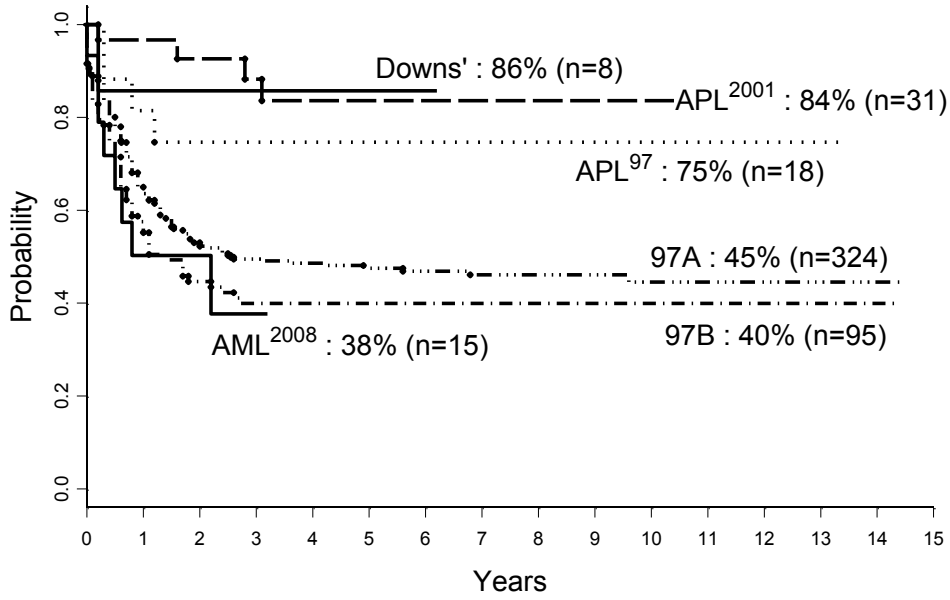


Fig 6. TPOG AML97. Event-free survival by protocol. (1997.1~2011.8)

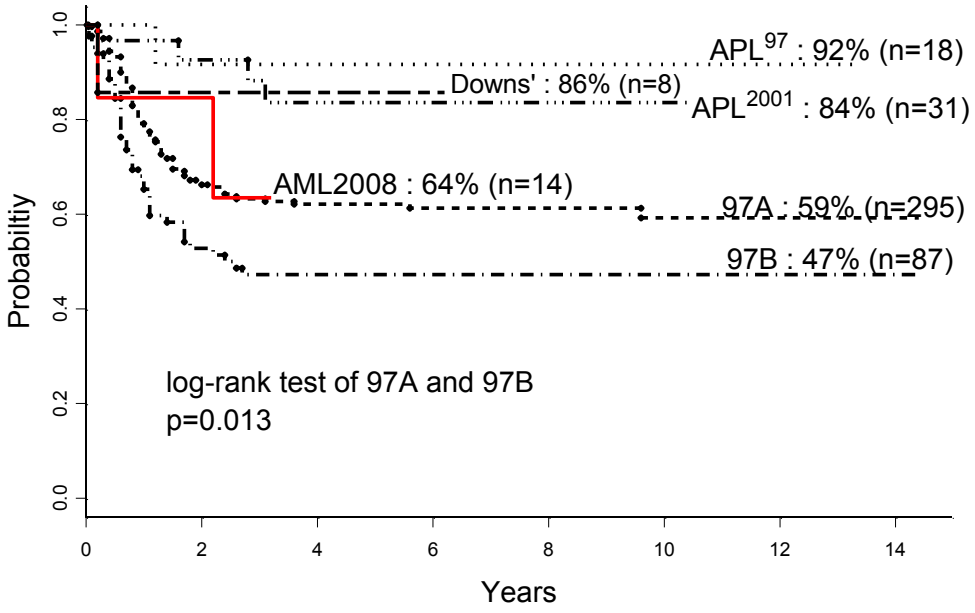
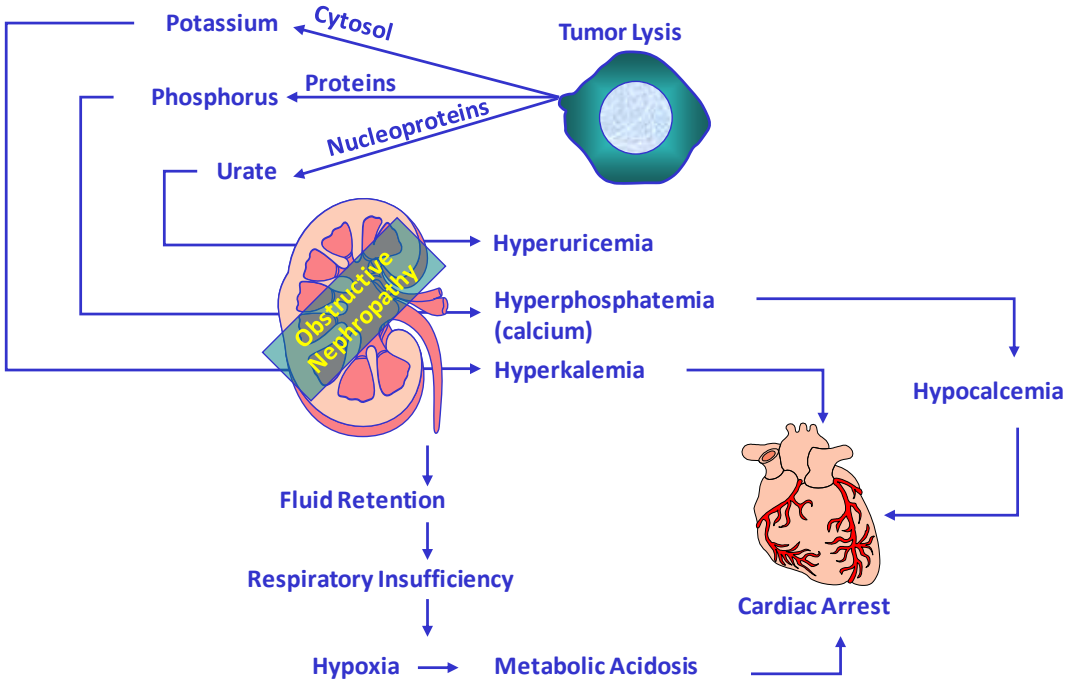


Fig 7. TPOG AML97. Disease-free survival by protocol. (1997.1~2011.8)

Consequences of tumor lysis syndrome



Administration of Antihyperuricemic Agents

Agent	Recommendation
Allopurinol	Dosing: 100 mg/m^2/dose every 8 hours (10 mg/kg/d divided every 8 hours) PO (maximum, 800 mg/d) or 200-400 mg/m^2/d in 1-3 divided doses; IV (maximum, 600 mg/d) Reduce dose by 50% or more in renal failure Reduce 6-mercaptopurine and/or azathioprine doses by 65%–75% with concomitant allopurinol May need to adjust doses of dicumarol, thiazide diuretics, chlorpropamide, cyclosporine, or allopurinol when they are used concomitantly with allopurinol
Rasburicase	Contraindicated in glucose-6-phosphate dehydrogenase–deficient patients, as well as in patients with a known history of anaphylaxis or hypersensitivity reactions, hemolytic reactions, or methemoglobinemia reactions to rasburicase or any of the excipients Administration: intravenously over 30 minutes according to dosages recommended in Table 8 Uric acid levels should be monitored regularly and used as a guide to modulate dosing; to measure uric acid levels place blood sample immediately on ice to avoid continual pharmacologic ex vivo enzymatic degradation 10% incidence of antibody formation

Management of tumor lysis syndrome

1. Hydration (3000ml/m2/day), diuresis
2. Correction of metabolic abnormalities (hyperphosphatemia, hyperkalemia, hypocalcemia)
3. Hypouricemic agents (allopurinol and rasburicase)
4. Hemofiltration or hemodialysis

Management of metabolic abnormalities in tumor lysis syndrome

Abnormality	Management Recommendation
Hyperphosphatemia	
Moderate, ≥ 2.1 mmol/L	Avoid IV phosphate administration Administration of phosphate binder
Severe	Dialysis, CAVH, CVVH, CAVHD, or CVVHD
Hypocalcemia, ≤ 1.75 mmol/L	
Asymptomatic	No therapy
Symptomatic	Calcium gluconate 50-100 mg/kg IV administered slowly with ECG monitoring
Hyperkalemia	
Moderate and asymptomatic, ≥ 6.0 mmol/L	Avoid IV and oral potassium ECG and cardiac rhythm monitoring Sodium polystyrene sulphonate
Severe (> 7.0 mmol/L) and/or symptomatic	Same as above, plus: Calcium gluconate 100-200 mg/kg by slow IV infusion for life-threatening arrhythmias Regular insulin (0.1 U/kg IV) + D25 (2 mL/kg) IV Sodium bicarbonate (1-2 mEq/kg IV push) can be given to induce influx of potassium into cells. However, sodium bicarbonate and calcium should not be administered through the same line. Dialysis
Renal dysfunction (uremia)	Fluid and electrolyte management Uric acid and phosphate management Adjust renally excreted drug doses Dialysis (hemo- or peritoneal) Hemofiltration (CAVH, CVVH, CAVHD, or CVVHD)

常用化學藥物及副作用

藥名	給藥途徑	副作用
Vincristine (Oncovin) (透明小針)	靜脈注射	噁心、嘔吐、骨髓功能受抑制、靜脈炎、掉頭髮、**腹痛、腸蠕動不良(便祕)、神經毒性(排尿困難、暫時性跛行、手腳麻刺感)** *不可外漏
Epirubicin (小紅莓)	靜脈注射 靜脈點滴注射	噁心、嘔吐、**心臟毒性**、黏膜潰瘍、**尿液顏色改變** *不可外漏
Idarubicin (小紅莓)	靜脈注射 靜脈點滴注射	噁心、嘔吐、口腔黏膜潰瘍、腹瀉、骨髓抑制、肝臟毒性、掉頭髮、**心臟毒性** *不可外漏
Asparaginase (Leunase) (屁股針)	肌肉注射 靜脈注射	噁心、嘔吐、發燒、**蕁麻疹(過敏)、胰臟功能障礙、高血糖**、骨髓功能受抑制、**局部紅腫、凝血及出血功能異常**
Methotrexate (MTX) (黃藥)	口服 肌肉注射 靜脈點滴注射 脊髓腔注射	噁心、嘔吐、腹瀉、骨髓功能受抑制、掉頭髮、**口腔黏膜潰瘍、肝臟毒性、腎臟毒性、皮膚色素沉著、皮膚癢**
Cyclophosphamide (Endoxan) (膀胱針)	口服 靜脈注射 靜脈點滴注射	**噁心、嘔吐**、食慾不振、骨髓功能受抑制、**出血性膀胱炎**、口腔黏膜潰瘍、掉頭髮、皮膚色素沉著、腎臟毒性
Ifosfamide (膀胱針)	靜脈注射 靜脈點滴注射	噁心、嘔吐、骨髓功能受抑制、掉頭髮、出血性膀胱炎、腎臟毒害
Etoposide (VP-16) (泡泡針)	靜脈點滴注射	噁心、嘔吐、食慾不振、頭痛、喘息、骨髓功能受抑制、掉頭髮、口腔黏膜潰瘍、肝、腎毒性高、**繼發性腫瘤的形成**
Cytarabine (Ara-C)	皮下注射 靜脈點滴注射 靜脈注射 脊髓腔注射	噁心、嘔吐、**口腔黏膜潰瘍、腹瀉、骨髓功能受抑制**、肝臟毒性、肺臟毒性、掉頭髮、**皮膚疹、結膜炎**、靜脈炎、**發燒**
Mercaptopurine (6-MP)	口服	噁心、嘔吐、食慾不振、骨髓功能受抑制、腹瀉、肝臟毒性
Mitoxantrone (Novantrone) (小藍莓)	靜脈點滴注射	噁心、嘔吐、食慾不振、骨髓功能受抑制、掉頭髮、肝臟毒性、口腔黏膜潰瘍 *不可外漏

Cisplatin (白金)	靜脈點滴注射	**嚴重噁心及嘔吐**、食慾不振、骨髓功能受抑制、**嚴重的腎毒性、耳鳴或聽力喪失**、皮膚色素沉著、末梢神經病變(手、腳麻木感)、**電解質不平衡**
Carboplatin (第二代白金)	靜脈點滴注射	噁心、嘔吐、**骨髓功能受抑制**、腎臟毒性
Dactinomycin (Actinomycin-D)	靜脈注射 靜脈點滴注射	**噁心、嘔吐**、骨髓功能受抑制、掉頭髮、腸胃黏膜潰瘍及發炎、腹瀉、皮膚色素沉著、毛囊性青春痘、靜脈炎 *不可外漏
Doxorubicin (Adriamycin)	靜脈注射 靜脈點滴注射	噁心、嘔吐、心臟毒性、骨髓功能受抑制、嚴重掉髮、肝臟毒性、皮膚色素沉著、靜脈炎、黏膜潰瘍、尿液顏色改變 *不可外漏
Bleomycin	靜脈注射	噁心、嘔吐、口腔炎、食慾不振、**肺部纖維化**、肺炎、皮膚色素沉著、掉頭髮、發燒
Busulfan	口服 靜脈點滴注射	噁心、嘔吐、**骨髓功能受抑制**、肺臟毒性、皮膚色素沉著、掉頭髮、**抽搐**
Dacarbazine (DTIC)	靜脈注射	噁心、嘔吐、食慾不振、靜脈炎、顏面潮紅、骨髓功能受抑制、「似流行性感冒」的症狀、肝臟毒性、掉頭髮、注射部位疼痛
Hydroxyurea	口服	噁心、嘔吐、皮膚紅疹、骨髓功能受抑制、掉頭髮、口腔黏膜潰瘍、頭痛
Topotecan	靜脈點滴注射	噁心、嘔吐、食慾不振、**嚴重骨髓功能受抑制**、腹痛、頭痛

兒童血液科　住院病人化療止吐藥物開方原則　2016/6/22

（因應 JCI 評鑑感控原則，ampoule 藥物不留餘量）

Ondansetron (Zofran) 8mg/amp　健保給付價 $200/1PC　半衰期 2.8 hours

0.2 mg/kg iv slowly push started 30mins before chemotherapy

Q8H for high or moderate emetogenic regimen （請開 3PC/day 回病房）

QD for low emetogenic regimen

開立單位為整數（無條件進位），max single dose: 8mg

Granisetron (Kytril) 3mg/amp　健保給付價 $674/1PC　半衰期 5 ~ 9 hours

0.04 mg/kg iv slowly push started 30mins before chemotherapy

QD use for highly or moderate emetogenic regimen

開立單位起跳為 1mg，單位為 0.5mg，建議 20kg 以上使用

60kg 以上直接 3mg

Palonosetron (Aloxi) 0.25mg/vial　健保給付價 $861/1PC　半衰期 29.5 hours

（仿單無兒童適應症，建議使用於青春期以後病人）

0.25mg iv slowly push started 30mins before chemotherapy **QOD** use

高致吐性 regimen 考慮同時併用 Dexan 0.15mg/kg Q12H (max 5mg)

若 breakthrough emesis 或非化療期間之止吐藥物，癌症病人可考慮基金會用藥，

（兒癌）Zofran 4mg/amp　或　（兒癌）Kytril 1mg/amp

（兒癌）Zofran 8mg/tab　或　（兒癌）Kytril 1mg/tab

Antineoplastic Agents with HIGH Emetic Risk
> 90% frequency of emesis in absence of prophylaxis

Single agent antineoplastic therapy

Altretamine	Dactinomycin
Carboplatin	Mechlorethamine
Carmustine > 250 mg/m^2	Methotrexate ≥ 12 g/m^2
Cisplatin	Procarbazine (oral)
Cyclophosphamide ≥1 g/m^2	Streptozocin
Cytarabine 3 g/m^2/dose	Thiotepa ≥ 300 mg/m^2
Dacarbazine	

Multiple agent antineoplastic therapy

With the exceptions listed below, emetogenicity is classified based on the most highly emetogenic agent.

The following are also classified as high emetic risk:

Cyclophosphamide + anthracycline
Cyclophosphamide + doxorubicin
Cyclophosphamide + epirubicin
Cyclophosphamide + etoposide
Cytarabine 150-200 mg/m^2 + daunorubicin
Cytarabine 300 mg/m^2 + etoposide
Cytarabine 300 mg/m^2 + teniposide
Doxorubicin + ifosfamide
Doxorubicin + methotrexate 5 g/m^2
Etoposide + ifosfamide

Antineoplastic Agents with LOW Emetic Risk
10% to <30% frequency of emesis in absence of prophylaxis

Single agent antineoplastic therapy

Cytarabine <200mg/m^2
Etoposide
Mitoxantrone
Topotecan

Antineoplastic Agents with MODERATE Emetic Risk
30-90% frequency of emesis in absence of prophylaxis

Single agent antineoplastic therapy

Aldesleukin > 12 to 15 million units/m^2
Amifostine > 300 mg/m^2
Arsenic trioxide
Azacitidine
Bendamustine
Busulfan
Carmustine ≤ 250 mg/m^2
Clofarabine
Cyclophosphamide < 1 g/m^2
Cyclophosphamide (oral)
Cytarabine > 200 mg to < 3 g/m^2
Daunorubicin
Doxorubicin
Epirubicin
Etoposide (oral)
Idarubicin
Ifosfamide
Imatinib (oral)
Intrathecal therapy
(methotrexate, hydrocortisone & cytarabine)
Irinotecan
Lomustine
Melphalan > 50 mg/m^2
Methotrexate ≥ 250 mg to < 12 g/m^2
Oxaliplatin > 75 mg/m^2
Temozolomide (oral)
Vinorelbine (oral)

Criteria for Staging Acute Graft-Versus-Host Disease

Stage	Skin	Liver	Intestinal Tract
1	**Maculopapular rash < 25% of body surface**	Bil 2-3 mg/dL Bil < 2 mg/dL with Alk-p 2 x nl and liver Bx consistent with GVHD	> 500 mL diarrhea/day (300 mL/M^2/day) Nausea; anorexia
2	**Maculopapular rash 25-50% of body surface**	Bil 3-6 mg/dL	1000 mL diarrhea/day (>600 mL/M^2/day) Vomiting; food intolerance
3	Generalized erythroderma	Bil 6-15 mg/dL	> 1500 mL diarrhea/day (>1000 mL/M^2/day)
4	**Generalized erythroderma with bullous formation and desquamation**	**Bil > 15 mg/dL**	**Severe abd. Pain, with or without ileus**

治療分期(**Grading for Treatment Criteria)**

a. 輕度 GVHD = 僅侵犯皮膚第 I 或 II 期

b. 中度 GVHD = 皮膚第 I 至 III 期合併肝臟第 I 至 III 期或腸胃道第 I 至 III 期

c. 重度 GVHD =皮膚、肝臟或腸胃道第 IV 期

（以上摘錄自兒科治療手冊）

6L 常用針劑型抗微生物製劑整理

1. Cefazoline 100mg/kg/day iv q6h or q8h
2. Cefuroxime (Zinacef) 100mg/kg/day iv q6-8h
3. Ceftazidime (Fortum) 100-150mg/kg/day IVF q6-8h
4. Cefepime 50mg/kg/dose IVF q12h
5. Gentamycin 2mg/kg/dose IVF q8h or 5mg/kg/dose IVF qd
6. Amikacin 5mg/kg/dose IVF q8h
7. Tienam 15-20mg/kg/dose IVF q6h
8. Meropenem 15-20mg/kg/dose IVF q8h
9. Vancomycin 10~15mg/kg/dose IVF q6~8h (max. 60mg/kg/day)
10. Teicoplanin 10 mg/kg IVF q12h × 3 doses, then 6 mg/kg IVF qd (10mg/kg if severe infection)
11. Augmentin 100-120mg/kg/day IVF q6-8h
12. Tazocin 100mg/kg/dose IVF q8h
13. Fluconazole 5-12mg/kg/day IVF qd
14. Voriconazole 6mg/kg/dose IVF q12h x 1 day, then 4mg/kg/dose IVF q12h
15. Posaconazole 13 歲以上, 5cc TID for prophylaxis, 10cc BID for treatment; or 3 tab QD; 小於 13 歲請與主治確認劑量給法及使用時機
16. Caspofungin 75mg/m2/dose IVF qd x 1 day, then 50mg/m2/dose IVF qd
17. Micafungin 1mg/kg QD for prophylaxis (max 50mg), 2mg/kg QD for treatment (max 100mg)
18. Acyclovir 10mg/kg/dose IVF q8h; 500mg/m2/dose IVF q8h for herpes zoster

表 1. 中國兒童各年齡層正常紅血球各項值

Group No.	Age	No. of cases	RBC ($10^6/mm^3$)	Hgb (g/dl)	Hct (%)	MCV (fl)	MCH (pg)	MCHC (%)
1	1/2-1d	28	*4.47±0.46	15.7±1.5	47.8±4.5	107±3.1	35.1±1.2	32.9±0.4
2	1-3d	34	4.52±0.46	15.7±1.7	46.6±4.8	103±3.7	34.7±1.3	33.7±0.8
3	3-5d	32	4.41±0.41	15.5±1.5	45.7±4.2	103±3.6	34.9±1.1	33.9±0.8
4	5-8d	21	4.42±0.43	15.3±1.3	44.8±3.8	101±3.6	34.7±1.4	34.1±0.8
5	1mo	24	3.41±0.52	11.0±1.4	31.9±4.6	95±3.8	32.3±1.4	34.1±0.8
6	2mo	36	3.47±0.23	10.4±0.7	30.6±2.1	88±4.1	30.0±1.6	34.0±1.1
7	2-4mo	31	4.14±0.48	11.6±1.1	33.6±3.1	81±4.9	27.9±1.7	34.5±1.1
8	4-6mo	32	4.44±0.39	11.8±1.0	34.6±2.6	78±3.1	26.7±1.2	34.2±0.8
9	6-12mo	40	4.70±0.27	12.5±0.8	36.6±2.1	78±2.4	26.5±0.8	34.1±1.0
10	1-2yr	49	4.68±0.35	12.4±0.8	36.6±2.4	78±3.1	26.6±1.4	34.0±0.9
11	2-4yr	43	4.68±0.37	12.6±0.8	37.1±2.7	79±2.9	26.9±1.2	34.0±1.1
12	4-6yr	54	4.62±0.34	12.7±1.0	37.5±2.5	81±3.6	27.5±1.6	33.8±1.2
13	6-7yr	44	4.76±0.35	13.4±0.9	39.0±2.5	82±2.9	28.1±1.3	34.2±1.2
14	7-8yr	40	4.70±0.37	13.2±0.9	38.8±2.6	83±3.6	28.2±1.3	34.1±1.2
15	8-9yr	38	4.66±0.30	13.4±0.8	38.7±2.3	83±3.1	28.6±1.5	34.4±1.1
16	9-10yr	30	4.86±0.35	13.8±0.9	41.0±2.7	85±2.9	28.4±1.2	33.7±0.9
17	10-11yr	34	4.74±0.30	13.4±0.6	40.5±2.3	86±3.8	28.3±1.1	33.0±1.2
18	11-12yr	38	4.66±0.30	13.3±0.8	40.1±3.3	86±3.8	28.5±1.1	33.2±1.4
Adults	Male	55	5.30±0.40	15.5±1.3	47.0±2.5	89±6.3	29.0±2.5	33.0±1.0
	Female	52	4.50±0.40	13.0±1.0	39.5±2.3			

* Mean±ISD ** Adult samples were tested on the same model of counter (Sysmex CC-170) in 1983.

（上表摘錄自兒科治療手冊）

表 2. 中國兒童各年齡層白血球數及分類

Group No.	Age	No. of cases	WBC ($10^6/mm^3$)	Neutrophils (%)	Lymphocytes (%)	Monocytes (%)	Eosinophils (%)
1	1/2-1d	29	*19.3±5.6	70.4±9.2	20.3±7.4	6.5±3.0	1.6±1.7
2	1-3d	36	10.7±2.4	60.4±10.3	26.5±9.7	9.0±3.8	3.3±2.8
3	3-5d	32	8.3±3.1	51.9±12.3	29.6±11.9	14.7±4.3	2.3±1.7
4	5-8d	21	10.1±3.3	40.4±7.8	40.5±7.6	14.3±5.3	2.7±1.9
5	1mo	27	8.0±1.5	21.0±8.7	65.7±10.1	8.0±3.7	4.9±2.4
6	2mo	37	7.7±2.4	17.2±7.8	72.4±9.6	6.1±3.2	3.8±2.9
7	2-4mo	36	8.3±2.7	20.6±9.8	71.0±10.7	6.1±4.2	2.1±1.9
8	4-6mo	34	9.6±2.4	20.0±7.6	71.8±9.0	6.0±3.5	2.3±2.2
9	6-12mo	44	9.1±2.6	21.8±6.9	69.7±12.9	4.2±3.0	2.3±1.5
10	1-2yr	52	9.1±2.7	27.3±13.4	66.3±15.3	4.3±2.9	1.7±1.6
11	2-4yr	44	8.9±2.2	36.1±13.4	57.4±12.7	3.8±2.5	2.1±2.2
12	4-6yr	53	8.2±2.2	42.7±14.9	48.6±14.2	4.8±2.8	3.3±3.5
13	6-7yr	47	7.4±2.2	45.6±12.0	45.5±12.5	4.3±2.5	3.9±3.4
14	7-8yr	40	7.5±2.0	49.3±11.7	44.1±11.7	4.1±2.2	2.4±2.4
15	8-9yr	40	7.6±1.4	50.2±11.9	42.2±11.4	4.0±2.0	3.3±3.1
16	9-10yr	29	7.9±1.5	52.9±8.4	39.6±8.9	3.5±2.1	3.1±2.5
17	10-11yr	36	6.9±1.5	45.5±10.9	74.4±12.4	4.2±2.4	3.0±2.9
18	11-12yr	40	6.5±1.4	48.6±10.5	43.5±9.3	4.9±2.7	2.7±2.4

* Mean±ISD

（上表摘錄自兒科治療手冊）

Guidelines for pediatric RBC transfusions (Nelson Textbook of Pediatrics, 19th Edition)

CHILDREN AND ADOLESCENTS
Acute loss of > 25% of circulating blood volume
Hemoglobin < 8.0 g/dL in the perioperative period
Hemoglobin < 13.0 g/dL and severe cardiopulmonary disease
Hemoglobin < 8.0 g/dL and symptomatic chronic anemia
Hemoglobin < 8.0 g/dL and marrow failure
INFANTS ≤ 4 MO OLD
Hemoglobin < 13.0 g/dL and severe pulmonary disease
Hemoglobin < 10.0 g/dL and moderate pulmonary disease
Hemoglobin < 13.0 g/dL and severe cardiac disease
Hemoglobin < 10.0 g/dL and major surgery
Hemoglobin < 8.0 g/dL and symptomatic anemia

Guidelines for pediatric PLT transfusions (Nelson Textbook of Pediatrics, 19th Edition)

CHILDREN AND ADOLESCENTS
PLT count < 50×10^9/L and bleeding
PLT count < 50×10^9/L and an invasive procedure
PLT count < 20×10^9/L and marrow failure with hemorrhagic risk factors
PLT count < 10×10^9/L and marrow failure without hemorrhagic risk factors
PLT count at any level, but with PLT dysfunction plus bleeding or an invasive procedure
INFANTS ≤ 4 MO OLD
PLT count < 100×10^9/L and bleeding or during extracorporeal membrane oxygenation
PLT count < 50×10^9/L and an invasive procedure
PLT count < 20×10^9/L and clinically stable
PLT count < 50×10^9/L and clinically unstable
PLT count at any level, but with PLT dysfunction plus bleeding or an invasive procedure

Guidelines for pediatric plasma transfusions (Nelson Textbook of Pediatrics, 19th Edition)

INFANTS, CHILDREN, AND ADOLESCENTS
Severe clotting factor deficiency AND bleeding
Severe clotting factor deficiency and an invasive procedure
Emergency reversal of warfarin effects
Dilutional coagulopathy and bleeding (e.g., massive transfusion)
Anticoagulant protein (antithrombin III, proteins C and S) replacement
Plasma exchange replacement fluid for thrombotic thrombocytopenic purpura or for disorders in which there is risk of bleeding due to clotting protein abnormalities (e.g., liver failure)

* Words in italics must be defined for local transfusion guidelines.

Table 1. Guidelines for transfusion of WB or reconstituted WB

1. Exchange transfusion for:
   - HDN
   - *Hyperbilirubinemia* with risk of kernicterus
2. After cardiopulmonary bypass
3. ECMO
4. Massive transfusion*

*Defined as transfusion of >1 blood volume in 24 hr.

Table 2. Guidelines for transfusion of RBCs in patients less than four months of age

1. Hct <20% with *low reticulocyte* count and symptoms of anemia*
2. Hct <30% with an infant:
   - On <35% hood O2
   - On O2 by nasal cannula
   - On continuous positive airway pressure and/or intermittent mandatory ventilation with mechanical ventilation with mean airway pressure <6 cm H2O
   - With *significant* apnea or bradycardia†
   - With *significant* tachycardia or tachypnea‡
   - With low weight gain§
3. Hct <35% with an infant:
   - On >35% hood O2
   - On continuous positive airway pressure/intermittent mandatory ventilation with mean airway pressure ≥6-8 cm H2O
4. Hct <45% with an infant:
   - On ECMO
   - With congenital cyanotic heart disease

*Tachycardia, tachypnea, poor feeding.

†More than six episodes in 12 hr or two episodes in 24 hr requiring bag and mask ventilation while receiving therapeutic doses of methylxanthines.

‡Heart rate >180 beats/min for 24 hr; respiratory rate >80 breaths/min for 24 hr.

§ Gain of <10 g/day observed over 4 days while receiving ≥100 kcal/kg/day.

Table 3. Guidelines for transfusion of RBCS in patients more than four months of age

1. Emergency surgical procedure in patient with *significant preoperative anemia*
2. Preoperative anemia when other corrective therapy is not available
3. Intraoperative blood loss ≥15% total blood volume
4. Hct <24%:
   - In perioperative period, with *signs and symptoms of anemia*
   - While on chemotherapy/radiotherapy
   - Chronic congenital or acquired *symptomatic* anemia
5. Acute blood loss with *hypovolemia* not responsive to other therapy
6. Hct <40% with:
   - Severe pulmonary disease
   - ECMO
7. Sickle cell disease:
   - Cerebrovascular accident
   - Acute chest syndrome
   - Splenic sequestration
   - Recurrent priapism
   - Preoperatively when general anesthesia is planned to reach Hb = 10 g/dL
8. Chronic transfusion programs for disorders of RBC production (such as β-thalassemia major and Diamond-Blackfan syndrome unresponsive to therapy)

Table 4. Guidelines for platelet transfusion in children

1. Platelet count 5-10 × 10^9/L with failure of platelet production
2. Platelet count <30 × 10^9/L in neonate with failure of platelet production
3. Platelet count <50 × 10^9/L in stable premature infant:
   - With *active bleeding*
   - Invasive procedure with failure of platelet production
4. Platelet count <100 × 10^9/L in sick premature infant:
   - With *active bleeding*
   - Invasive procedure in patient with DIC

Table 5. Guidelines for platelet transfusion in patients with a normal platelet count

1. Active bleeding in association with a qualitative platelet defect
2. Unexplained, *excessive bleeding* in a patient undergoing cardiopulmonary bypass
3. Patient undergoing ECMO:
   - With a platelet count of <100 × 10^9/L
   - With *higher platelet counts* and bleeding

Table 6. Guidelines for granulocyte transfusion in children

1. Neonates or children with *neutropenia* or granulocyte dysfunction with bacterial sepsis and lack of responsiveness to *standard therapy*
2. *Neutropenic* neonates or children with fungal disease not responsive to *standard therapy*

Table 7. Guidelines for the transfusion of FFP

1. Support during the management of disseminated intravascular coagulation
2. Replacement therapy:
   - When specific factor concentrates are not available, including but not limited to, antithrombin III, protein C or S deficiency, FII, FV, FX, and FXI
   - During therapeutic plasma exchange when FFP is indicated (cryo-poor plasma, plasma from which the cryoprecipitate has been removed, or S/D plasma may be beneficial in thrombotic thrombocytopenic purpura not responsive to conventional plasma exchange)
3. Reversal of coumadin in an emergency situation, such as before an invasive procedure with *active bleeding*

*FFP is not indicated for volume expansion or enhancement of wound healing

Table 8. Guidelines for the use of cryoprecipitate

1. *Hypofibrinogenemia* or dysfibrinogenemia with active bleeding.
2. *Hypofibrinogenemia* or dysfibrinogenemia, undergoing an invasive procedure
3. FXIII deficiency with active bleeding or undergoing an invasive procedure in the absence of FXIII concentrate
4. Limited directed donor cryoprecipitate for bleeding episodes in small children with hemophilia A (Note: previously untreated children should receive recombinant FVIII)
5. In the preparation of fibrin sealant
6. vWD: Active bleeding
   - Before an invasive procedure
   - Cryoprecipitate is only used in vWD disease when:
     - Deamino-D-arginine vasopressin (vasopressin) is contraindicated, not available or does not elicit response
     - Virally inactivated plasma-derived FVIII concentrate, which contains vWF, is not available

Table 10. Guidelines for the use of IVIG

1. Immune deficiency states—humoral
   - Primary, excluding patients with selective IgA deficiency
   - Secondary, including HIV infection, marrow transplant recipients
2. Hematologic diseases such as ITP, Evans syndrome, autoimmune hemolytic anemia (steroid resistant)
3. Prophylaxis to prevent opportunistic infection in marrow transplant patients
4. Sepsis neonatorum
5. Neonatal alloimmune thrombocytopenia
6. *Thrombocytopenia* secondary to maternal autoimmune disease (ITP, systemic lupus erythematosus)
7. Miscellaneous: Kawasaki syndrome, Guillain-Barré syndrome

Table 11. Guidelines for WBC-reduced blood components

1. Prevent recurrence of febrile nonhemolytic transfusion reactions
2. Reduce the risk of HLA alloimmunization
3. To prevent CMV transmission (see text)
4. Prevent post cardiopulmonary bypass lung injury

Table 13. Patients requiring irradiated blood components

1. Intra-uterine transfusion
2. Premature infants weighing <1200 g at birth
3. Patients with known or suspected cellular immune deficiencies
4. Patients undergoing marrow or peripheral blood progenitor cell transplant
5. Patients rendered *immunosuppressed* by chemotherapy or radiation treatment
6. Recipient of components from blood relatives
7. Recipients of HLA-matched or platelet crossmatch-compatible components

血液成份之種類及主要用途

血品名稱	內含有效成份	容積	用途	說明
紅血球濃厚液 Packed RBC	紅血球100ml;血漿40ml; 安定凝血因子40單位 ；白血球及血小板無作用	150ml	血液容積正常貧血病人之治療	絕大多數輸血應用此製品,盡量少用全血
冷凍紅血球 Frozen RBC	紅血球80ml; 缺血漿及凝血因子; 缺白血球及血小板	120 ml	用於因血漿蛋白抗體或白血球抗體而引起之發燒或過敏性輸血反應病人; 可用以儲存稀少血型	冷凍紅血球製備後應於24小時之內輸用,製備後不能退回血庫.
乏白血球性紅血球 Leukocyte poor RBC	紅血球90ml; 缺白血球及血小板; 有些製品缺血漿及凝血因子	120 ~ 150 ml	用於因白血球抗體而引起之發燒反應病人, 或用於長期輸血之病人	乏白血球性紅血球應於24小時之內輸用, 製備後不能退回血庫.
全血 Whole blood, stored	紅血球100ml;血漿150ml;安定凝血因子120單位; 白血球及血小板無作用, 缺不穩定凝血因子	285 ml	大量失血時補充用 (紅血球及血液容積均減少時)	全血應只用於補充血紅素及血量均不足之病人; 要補充血小板及白血球. 應使用血小板或白血球濃厚液.
新鮮全血,儲存少於三天 Whole blood, fresh	紅血球100ml;血漿150ml;安定凝血因子120單位; 少量不穩定凝血因子80 單位, 少量白血球, 及血小板1單位	285 ml	新生兒或嬰兒或病重病人補充紅血球及血液容積用, 不適用為補充血小板及白血球或凝血因子之用 (量太少)	新鮮全血所含的血小板及白血球量有限.

血品名稱	內含有效成份	容積	用途	說明
冷凍血漿 Stored frozen plasma	血漿120ml; 安定凝血因子90單位; 缺 V及VIII凝血因子, 無血小板.	100ml	安定凝血因子缺乏病人(如肝病病人, 缺II, VII, IX, X, XI等因子)之治療; 一般體液擴充白蛋白之用	除了V及VIII,一般凝血因子均可由冷凍血漿獲得(不必用新鮮血漿)解凍後不能退回血庫. 應於24小時之內輸用
新鮮冷凍血漿 Fresh frozen plasma	血漿120ml; 所有凝血因子100單位; 無血小板.	135ml	大量輸血病人補充凝血因子及體液擴充之用, (除了缺V, VIII因子外, 一般缺凝血因子之治療應使用冷凍血漿)	解凍後不能退回血庫;應於6小時之內輸用(以保持不穩定因子, 若在24小時之內可以當作冷凍血漿輸用).
冷凍沉澱品 Cryoprecipitate	纖維素原, 100mg; VIII及XIII因子, 50單位; fibronectin; von Willebrand因子, 50單位	20ml	治療VIII (A型血友病)及XIII因子缺乏之病人; 治療von Willebrand病人;治療纖維原過低血症	冷凍沉澱品解凍後, 不能退回血庫, 應於6小時之內輸用以保持不穩定凝血因子之活性.
血小板濃厚液 Platelet concentrate	血小板, 每袋大約有200至300億個血小板, (12單位之袋); 少量顆粒球	200ml	用於因血小板缺乏或血小板病變所引起之出血, (包括大量輸血病人補充血小板)	血小板濃厚液不一定要使用相同ABO血型;不使用微孔過濾器.

（上表轉製自孫建峰醫師所著『輸血醫學』）

輸血反應的臨床處置

程度(級)	反應症狀	可能之原因	處置
(I)	風疹,蕁麻疹	血漿蛋白之輕度過敏反應	降低輸注速度, 給與抗組織垵劑, 經過30分鐘後若沒有改變,當作 (II) 級處理.
(II)	顏面潮紅 蕁麻疹, 皮膚疹, 發冷, 發燒 煩躁不安 心跳加速	血漿蛋白之中度至重度過敏現象 對RBC,WBC及Platelet抗體之作用 血漿蛋白抗體之存在, 如IgA,細菌污染或熱性物質之存在	1. 停止血液之輸注, 但仍維持靜脈點滴. 2. 通知主治醫師及有關血庫 3. 將剩餘之血液連帶輸血套送往血庫,並抽取患者血液及尿送往檢查. 4. 給與抗組織垵(anti-histamine) 注射 (IM或IV), 或口服退熱劑 5. 若需要, 給與類固醇 (Corticosteroids). 6. 為預防再一次之反應,日後之輸血應使用特殊血液成分,如濃厚紅血球,洗滌紅血球,或去甘油之紅血球.
(III)	畏寒或寒戰 發燒, 不安 心跳加速 紅色小便 (血紅素尿症) 不明原因之出血, 血壓下降	Gram氏陰性菌血症 DIC 血管內溶血現象	1. 立即停止輸血, 維持靜脈鹽水點滴.之通暢 2. 通知主治醫師及血庫 3. 如 (II) 級之第3項處理 4. 休克之治療: 用血管收縮劑, 腎上腺素, 類固醇等, 並維持適當之靜脈輸液, 記錄患者之每日進出液量 (intake and output ect. fluid balance sheet) 5. 維持良好之腎臟血流, 如用furosemide 20-80mg IV 6. 預防DIC之爆發, 可用heparin 10,000 單位 IV 7. 若需要, 給與適當之相配合之血液成分. 8. 密切觀察病人,注意腎衰竭之徵兆,必要時施用腎透析.

(上表轉製自孫建峰醫師所著『輸血醫學』)

1) Packed RBC 或 Leukocyte-depleted RBC 之計算方式

預估所需紅血球量之方法

濃縮紅血球量之 C.C. 數 = 預計上升之 Hb 數 (gm/dl) x 3 x 體重 (公斤)

例如:

病人體重 20 公斤目前之血色素 (Hb) 為 6 gm/dl，若希望達到 Hb 10 gm/dl，則計算方式如下: Packed RBC = (10 - 6) x 3 x 20 = 240 c.c. (約 2 units of Packed RBC)

2) Leukocyte-depleted platelet concentrates 之計算方式: 小孩通常每 10 公斤輸 1 unit

預估所需濃縮血小板量之方法

以體表面積每平方公尺(BSA) 每 2 小單位血小板可上升大約 10,000 – 12,000 /mm^3/m^2

3) 血品血型次要選擇順序

病人血型																
	O				A				B				AB			
	血型選擇順序				血型選擇順序				血型選擇順序				血型選擇順序			
全血	O				A				B				AB			
紅血球	O				A		O		B		O		AB	O	A	B
血小板	O	A	B	AB	A	AB	O		B	AB	O		AB	A	B	
血漿	O	A	B	AB	A	AB			B	AB			AB			

(以上摘錄自兒科治療手冊)

ABO 血型不相合之移植病人輸血血型選擇

TABLE 25-2. Transfusion Support for Patients Undergoing ABO-Incompatible Allogeneic Hematopoietic Stem Cell Transplantation*

		Phase I	Phase II				Phase III
Recipient	Donor	All Components	Red Blood Cells	First Choice Platelets	Next Choice Platelets†	Fresh Frozen Plasma	All Components
O	A	Recipient	O	A	AB; B; O	A, AB	Donor
O	B	Recipient	O	B	AB; A; O	B, AB	Donor
O	AB	Recipient	O	AB	A; B; O	AB	Donor
A	O	Recipient	O	A	AB; B; O	A, AB	Donor
A	B	Recipient	O	AB	A; B; O	AB	Donor
A	AB	Recipient	A	AB	A; B; O	AB	Donor
B	O	Recipient	O	B	AB; A; O	B, AB	Donor
B	A	Recipient	O	AB	B; A; O	AB	Donor
B	AB	Recipient	B	AB	B; A; O	AB	Donor
AB	O	Recipient	O	AB	A; B; O	AB	Donor
AB	A	Recipient	A	AB	A; B; O	AB	Donor
AB	B	Recipient	B	AB	B; A; O	AB	Donor

*Modified with permission from Friedberg.[20]
†Platelet concentrates should be selected in the order presented.

◎ Phase I = from the time when the patient/recipient is prepared for marrow transplant/hematopoietic stem cell transplantation;

◎ Phase II = from the initiation of chemotherapy until 1) for Red Blood Cells, direct antiglobulin test is negative and antidonor isohemagglutinin is no longer detectable (ie, the back typing is donor type) or 2) for Fresh Frozen Plasma, recipient's erythrocytes are no longer detectable (ie, the front typing is consistent with donor's ABO type);

◎ Phase III = after the front and back type of the patient are consistent with donor's ABO type.

Replacement therapy for bleeding episodes in patients with hemophilia

	A 型血友病	B 型血友病
關節內出血	第一劑: 40u/kg, p.r.n: 20 u/kg on day 2, 3 類固醇 p.r.n	第一劑: 60-80 u/kg, p.r.n: 40 u/kg on day 2, 4
皮下血腫，拔牙 肌肉出血，鼻血	20u/kg, p.r.n QOD Transamin (口腔、鼻) 局部施壓	40 u/kg, p.r.n Q3d
嚴重口腔出血及 嚴重軟組織出血	40-50 u/kg	第一劑 80 u/kg →40 u/kg Q24h.會診血液科
後腹腔 (Iliopsoas hematoma)	50 u/kg→ 25 u/kg Q12h →20 u/kg QOD，共 10-14 天	80 u/kg→ 40 u/kg Q12-24hr, 會診血液科
血尿	休息，IVF→20 u/kg, 類固醇	30 u/kg（其它如左）
預防性	20-40 u/kg QOD	40 u/kg/Q3d

1. A 型血友病：每輸第 8 因子濃縮劑 1u/kg，可提昇血中濃度 2%。所需劑量依出血部位而定（上表）。嚴重出血、手術前，作侵入性檢查前或危及生命狀況時，需提昇至 100%以上。第一劑的 half life 較短，8 小時後應輸第 2 劑。之後的間隔為 12 小時。病人如需手術（牙、骨等），則在給第一劑 50u/kg 後連續以輸 3-4u/kg/hr。
2. B 型血友病：輸注血漿濃縮品{prothrombin complex concentrate（含 II、VII、IX、X）}或第九因子合成物。前者 1u/kg 可提昇血中濃度 1%，如選用後者則劑量需乘以 1.4 倍。
3. 輕度 A 型血友病出血時亦可試用 desmopressin。如缺 factor VIII 濃縮劑，可以 cryoprecipitate 取代（安全性較不確定）。
4. 手術前應做凝血蛋白抗體的篩檢。

（以上摘錄自兒科治療手冊）

癌症病人的嗜中性白血球低下發燒

(Neutropenic Fever in Cancer Patients)

兒童血液腫瘤科 江東和醫師

一、 簡介

Neutropenic fever (或稱 febrile neutropenia) 是 cancer patient 在化學治療過程中常見的 complication。Infection 是 neutropenic fever 最常見的原因，依國外的統計約佔 52%，其它造成 neutropenic fever 的原因包括 chemotherapy, blood transfusion, allergic reaction, 和 malignancy 本身。

二、 病理生理學

由於 cancer patient 在 neutropenic phase 若有 infections 可能有相當高的 mortality rate，因此所有的 neutropenic fever 病人均需當作嚴重感染的病人並給予 empirical antibiotic 治療。Bacteremia 是 neutropenic fever 常見的致病因， *Pseudomonas aeruginosa, E. Coli, Klebsiella pneumoniae* 是較常被 blood culture 培養出的致病菌，但其它革蘭氏陽性菌、或較難培養出的 anaerobes 和 fungus 也需常被考慮。 Lung, soft tissue (特別是 perirectal cellulitis) 或 mucosa 亦是好發感染的部位， UTI 或 CNS infection 在 neutropenic fever 的病人也是不少見的。

三、 臨床評估

(1) 病史：

有無接觸其它 Infection 的病人，特別是其它正在得水痘的病人。患者以前有無 documented sepsis 的病史以及最後一次化學治療的日期，在病人一到院時應先詢問清楚，以行推斷病人是否正處於 neutropenic stage。一般而言，化學治療後七至十天常是血球降到最低點 (Nadir) 的時候。

(2) 身體檢查：

任何看似微細無害的傷口 (innoculous-looking lesions) 均可以為 Sepsis 的病灶，故 oropharynx 有無傷口，手腳指有無 paronychia，肛門或外陰部有無發炎現象，病人最近有無靜脈注射的傷口或插導尿管。經驗上若病人有主訴腹痛、肛門疼痛或骨頭酸痛現象，其血液培養長出細菌的機率相當高。

(3) 實驗室檢查：

1. CBC, D/C, 生化及血液培養檢查為常規住院檢查，U/A 甚至 U/C 也應考慮作，因 neutropenic patient 可因 WBC count 低而無 pyuria 現象。
2. Neutropenic fever 病人若有 Paroxysmal cough, tachypnea, chest pain 等症狀應照 CXR。如皮膚有 Abscess 或傷口有 Pus formation 可以作 Pus culture。若有合併腹痛或腹瀉，加上病人有類似 Appendicitis 的症狀，應配合腹部 Plain film 或 Abdominal echo，以排除 typhlitis 的可能性。

3. 病人在使用抗生素治療時若又突然有一 spiking fever，考慮更換抗生素前應再作一次血液培養。

四、 檢驗與檢查

臨床上對 neutropenic fever 病人的診斷主要依賴血液培養的結果，尤其病人若距 neutropenia 恢復正常還有一段時間，任何的突然的 spiking fever 或 chills 皆可能另一次 septicemia，故應視情況再作 B/C 或 Fungus culture。

五、 處置及流程

(1) 對 Neutropenic fever 的治療必需先當作 Sepsis 處理，隨時留意病人是否有敗血症休克的情形，入院時儘快給予適當的抗生素及維持生命跡象的穩定。

(2) 若病人已有敗血症休克現象，則依敗血症休克原則處理，給予 normal saline，FFP 或 5% albumin。若有明顯血壓偏低症狀，應考慮給予 dopamine 來維持血壓，不要一味給予 fluid challenge。

(3) 在抗生素的選擇方面，neutropenic fever 病人若無發現其它病灶且生命跡象平穩，可先給予 cefamezine (100 mg/kg/D) + Gentamycin (2 mg/kg/dose) 。但病人若有 agraulocytosis 合併其它 toxic signs，可考慮優先使用 Ceftazidime (100 mg/kg/D) + Gentamycin or Amikacin (5 mg/kg/dose) 或 Cefepime 以 cover *Ps. aeruginosa*，必要時直接考慮使用 Carbopenem (Meropenem or Imipenem) 來 cover ESBL-producing strain 再視血液培養結果之 MIC 更改抗生素。

(4) 一旦燒退且白血球數目回昇後，一般在發燒完全消退三天以後，若無其它症狀可考慮先行出院。

(5) 若病人於住院期間有發現排便時肛門有明顯疼痛或紅腫，而發燒持續存在，需考慮加上 Metronidazole (20-40 mg/kg/D) 以 cover anaerobes 。若病人有裝 port-A or Hickman catheter，且表面皮膚有紅腫現象，必要時用 Vancomycin 或 Teicoplanin 代替 1st generation cephalosporin，俟三天內血液培養無 MRSA 之 isolation，再改回原來之 antibiotic。

(6) Neutropenic fever病人如發生以下情況，需考慮加上antifungal agents 如Fluconazole, Voriconazole 或 Caspofungin 等 (Echinocandins have fungicidal activity against Candida species and fungistatic activity against Aspergillus species).

1. 血液培養或皮膚切片有發現 Fungus
2. Endophalmitis
3. 使用廣效抗生素超過五天且血液培養尚未長出任何細菌
4. CXR 或 Water‘s view for sinus 有發現可疑類似黴菌感染之現象
5. 以前病史曾發生較侵襲性的黴菌感染
6. Abdominal echo 或 CT scan demonstrated multiple hepatosplenic microabscesses
7. 必要時考慮使用 G-CSF 縮短 neutropenic period

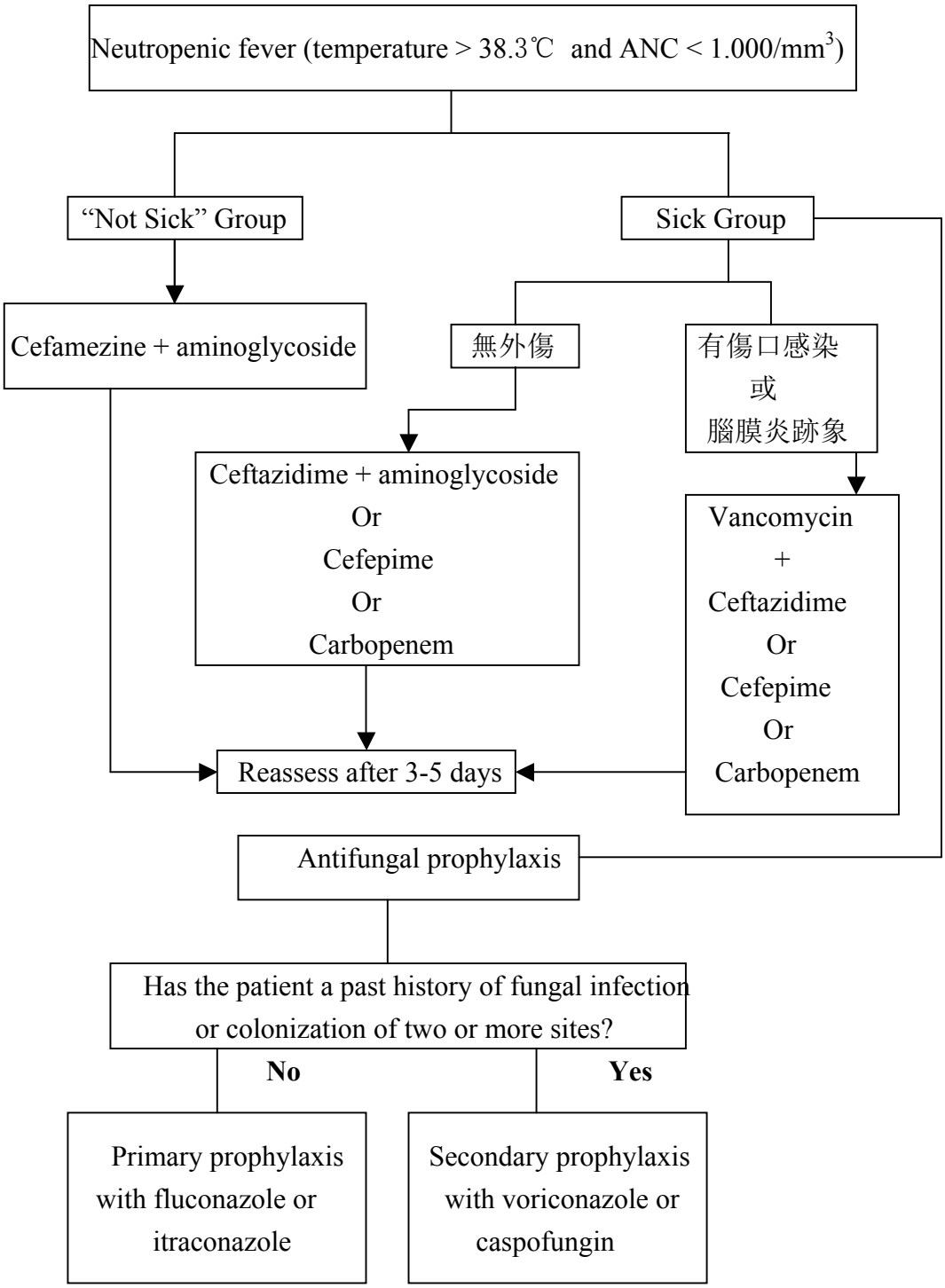
六、 預後

(1) 抽血驗 CBC, D/C, 生化, B/C 後，Set IV 時並先留備血以方便輸血。若病人的 vital sign 不穩定，應立即投予抗生素。

(2) 若病人有裝 port-A catheter，應優先從此處抽血並 Set IV，抗生素的使用也應 cover G(+)菌的感染。

(3) 如果病人病情嚴重或對目前使用的抗生素反應不佳，毋需等血液培養報告，考慮更換抗生素尤其須排除 ESBL-producing strains。

(4) 依據培養出的細菌藥物感受性調整抗生素。

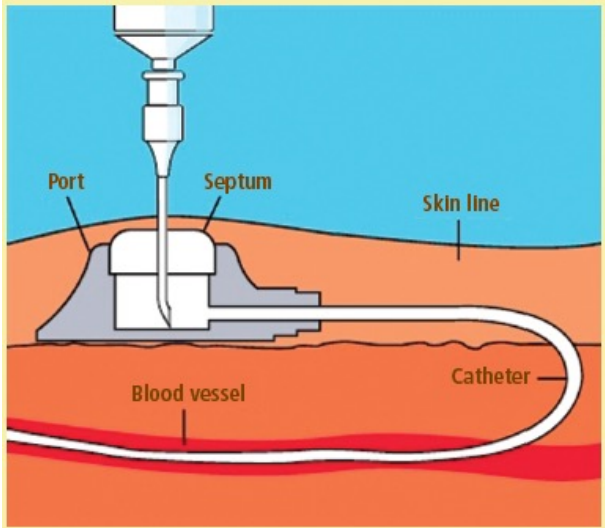


（此章節摘錄自兒科治療手冊）

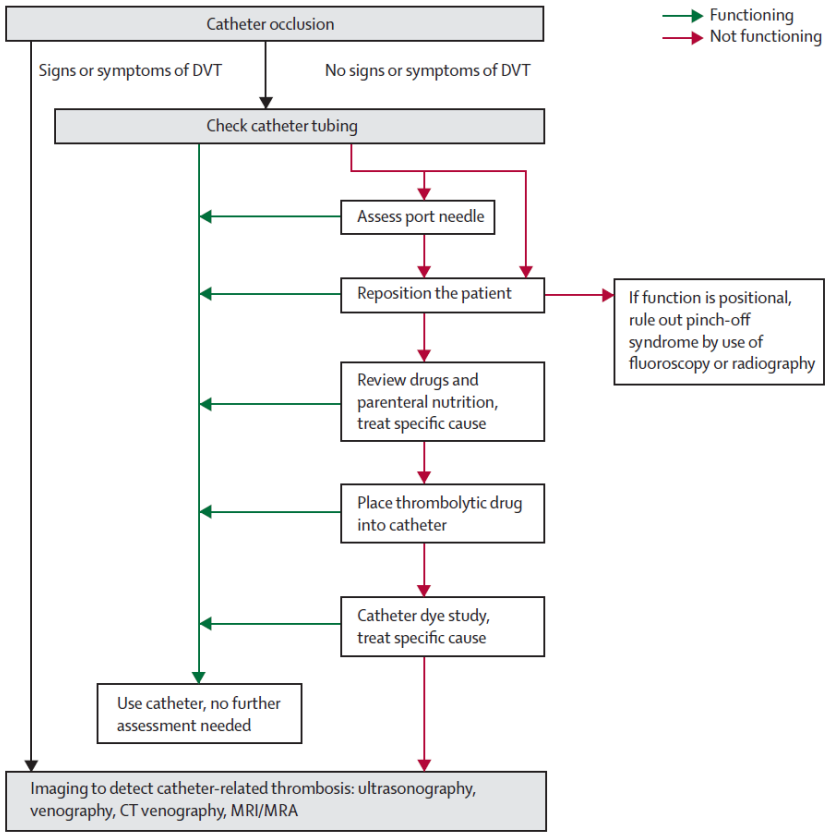
Guidelines for care and occlusion management of Port-A

Port A 使用注意事項

1. Port A 使用前須要 Port A needle 以直角角度進入皮下的 Port A device，接 normal saline，確定 push 及回抽皆順暢方可使用。
2. Port A needle 拔針時，需以稀釋之 heparin (本院藥品編號 PLC006P，500IU/10ml/vial) 一面 slowly push，一面 keep positive pressure 而後慢慢拔針， 以防血塊在 Port A 內形成。
3. Port A 可用於抽血，但須接上空 syringe， 先抽出 5ml-6ml 之血作廢， 再抽取所要之檢體。



Port-A 阻塞時的處理:



1. 先確定針頭位置、病人姿勢、以及目前使要的點滴藥物是否有問題
2. 考慮做靜脈攝影以確定血塊的存在與否， 血塊確定存在，可使用 Urokinase 5000u/lumen，打入 port-A 置放 小時。
3. 仍 順可再重複以上步驟一次，改為 Urokinase 10000u/lumen，置放 小時。
4. 以上的治 仍無法使人工血暢通，則可考慮手術移除。

給化學藥物過程須知及藥物外滲外漏的處理

戴上手套
↓
如用 IV lock, 需先連接 IV line (0.9% NS), 在遠端做 lock 之動作
在執行給化學藥前, 首先使用的 Syringe 含 0.9% saline, 確認回血通暢無阻.
↓
確定病人名字與給予的化學藥物無誤後, 才執行 iv slowly push 動作
當輸注化療藥之時，需隨時確認回血通暢及病人是否有任何不舒服
↓
拔針前，再 flush the line with 0.9% saline, 然後把 lock 鬆開

Prevention: 打藥前須先以 N/S push 確定回血通暢再打藥; 注射時, 如病人喊痛或注射有阻力, 立即停止注射, 並按照藥物外滲漏步驟處理, 打完拔針前以 N/S flush.

藥物外滲外漏的處理:
在 central venous access 相當普及的今天發生藥物外滲、外漏的情形普遍減少，但仍需熟悉外漏時的

緊急處理方法:

1. 停止注射或輸液。
2. 先　移除原　的注射管線及針頭, 用原針頭盡量反抽輸液。

i. 是滲　發生在週邊靜脈注射：在拔出針頭前先回抽注射管線內及外滲的液體。　是可取得解毒劑，除　皮下注射解毒劑之外，由原　的注射管線給予適當的解毒劑，可幫助解毒劑直接滲入化　藥品外滲的區域。

ii. 是滲　發生在中央靜脈注射：　是病人有植入implanted port 需評估是否　換針頭。回抽port pocket或導管內的液體。　可取得解毒劑則局部皮下注射解毒劑；　需靜脈注射解毒劑，注意過大的劑　會增加中央靜脈導管的壓　，應適　緩慢滴注。　病人有植入implanted port 在注射完解毒劑後應取出

3. 拔除針頭。

i. 外滲藥是**生物鹼**，如長春鹼Vinca類皮下注射Hyduronidase1500units及**溫水敷患處**以幫助吸收。

ii.其它具刺激性或**皮膚發泡性藥物-外滲部位以冰袋、冰枕或碎冰塊冰敷**，並用25號針投之空針皮下注射hydrocotisone 100mg或用dexamethasone 4mg於可能外滲部位的四周。

iii.外滲部位全部塗抹hydrocortisone乳膏。

4. 將患處抬高 48 小時
5. 冰敷：24 小時內 20 minutes q 4 h。不可熱敷 (唯 VP-16, vinblastine, vincristine 除外).
6. 患處用黃藥水敷 20 minutes q4h 直到紅腫減輕。
7. .避免在局部施壓。
8. 在 48 小時後鼓勵功能復健。
9. 如有嚴重疼痛、紅腫、或發生 ulcer 的早期，應立即會診手外科或整形外科。 大傷口是需要手術或補皮的。

Vesicant drugs	**Irritant drugs**	**Nonvesicant drugs**
Actinomycin D (Dactinomycin)	Carmustine	Asparanginase
Adriamycin(Doxorubicin)	cyclophosphamide	MTX
Daunomycin	etoposide (VP-16)	Ara-C
Idarubicin	ifosfamide	6-MP
Epirubicin	Dacarbazine（DTIC）	Carboplatin
Mitomycin-C	Bleomycin	
vincristin	Cisplatin	
vinblastine	Mitoxantrone	
Vinorelbine	Paclitaxel（Taxol）	

Vindesine	5-FU	

Antidote:

Drug	**Antidote**	**Compression**	**Onset of inflammatory**
Vincristine Vinblastin Etoposide Vinorelbine	Sodium chloride 0.9% 1ml or Hyaluronidase (150units/ml) 3~5ml by injection or Sodium bicarbonate (8.4%) 5ml or topical application ****禁用 Corticosteroids 及冷敷****	Hot 30~60min take rest 15min keep 1 day	12-24hrs
Epirubicin	Topical Dimethylsulfoxide (DMSO) 0r 50~90%, 15ml q2~4hr for 3day hydrocortisone 1pc(或用 dexamethasone4mg) + N/S 10ml	Cold 每4小時一次，每次15分,持續72小時或敷50分鐘休息10分鐘,持續24小時	1~2 weeks
Adramycin (Doxorubicin)	Topical Dimethylsulfoxide (DMSO) 0r 50~90%, 15ml q2~4hr for 3day hydrocortisone 1pc(或用 dexamethasone4mg) + N/S 10ml		
Daunorubicin	Topical Dimethylsulfoxide (DMSO) 0r 50~90%, 15ml q2~4hr for 3day hydrocortisone 1pc(或用 dexamethasone4mg) + N/S 10ml		
D-actinomycin	Sodium thiosulfate 10% (1.6ml in 3ml water)		

注意： 敷或熱敷時應避免壓迫到患部；熱敷溫 最好在攝氏 46–52 之間

HEMATOLOGIC EMERGENCY

F1 張從彥
2014.6.10

兒童血液腫瘤疾病急症之原因

- **腫瘤細胞產生之代謝產物**
- **本身腫瘤或是治療腫瘤後造成血球減少**
- **腫瘤壓迫到重要器官**

OUTLINE

- **Neutropenic fever**
- **Hyperleukocytosis**
- **Metabolic and electrolyte imbalance**
  - Tumor lysis syndrome
  - Hypercalcemia
  - SIADH
- **Anatomic region**
  - Superior cava syndrome
  - Abdominal emergency
  - Spinal cord compression
  - IICP & seizure

NEUTROPENIC FEVER

- **Risks:**
  - Underlying disease: leukemia, advanced lymphoma
  - Therapy: high dose chemotherapy, HSCT
  - Neutropenia: Absolute neutrophil counts (ANC) (=WBC * (Seg+Band %)) < 500
  - Disruptions of normal barrier
  - Nutrition, immunity, foreign body…

WORK-UP

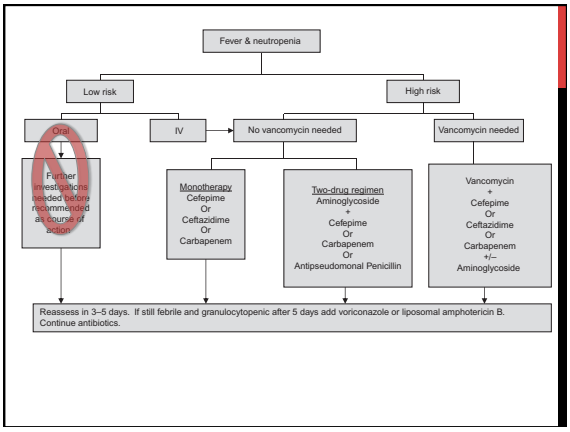
- **History & PE**
  - Oral cavity, pharynx, esophagus, lung, perineum and anus, whole skin
- **Complete blood counts, biochemistry, Blood culture, urine routine and culture**
- **Image study: CXR, Abdominal echography, CT**

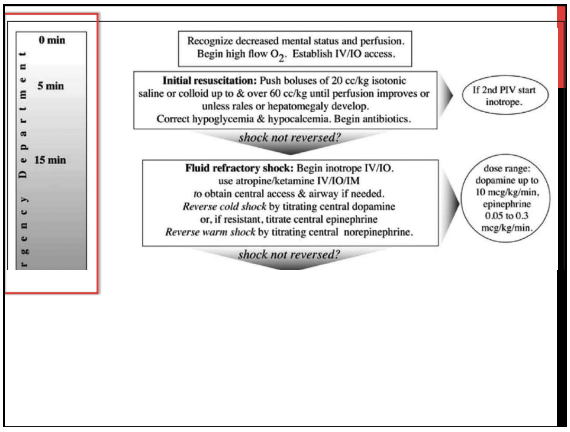
Low risk

ANC > 100
Normal chest radiograph
Expected resolution of neutropenia <10 days
Non toxic appearance
No comorbidities
Malignancy in remission

High risk

ANC < 100
Abnormal chest radiograph
Expected resolution of neutropenia >10 days
Toxic appearance
Presence of comorbidities
Not in remission





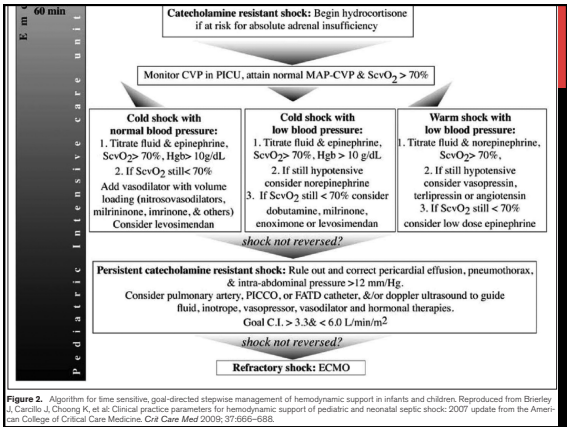


Figure 2. Algorithm for time sensitive, goal-directed stepwise management of hemodynamic support in infants and children. Reproduced from Brierley J, Carcillo J, Choong K, et al: Clinical practice parameters for hemodynamic support of pediatric and neonatal septic shock: 2007 update from the American College of Critical Care Medicine. *Crit Care Med* 2009; 37:666–688.

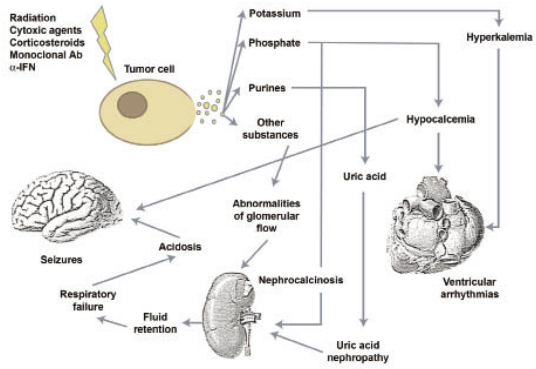
HYPERLEUKOCYTOSIS

- **定義：WBC > 100,000**
- **9~13% ALL, 5~22% AML**
- **Leukostasis: Myeloblast > 50,000, Lymphoblast > 100,000**
- **表現：**
  - CNS: blurred vision, confusion, somnolence, delirium, stupor, coma and papilledema
  - Pulmonary: tachypnea, dyspnea, hypoxia
  - Genitourinary: oliguria, anuria, priapism
  - Vascular symptoms which include DIC, retinal hemorrhage, myocardial infarction and renal vein thrombosis

- **治療：**
  - 減少白血球：leukapheresis (白血球分離術), exchange transfusion (換血) in infant, chemotherapy, hydroxyurea
  - Cautious blood transfusion
    - Keep Platelet > 20,000
    - Avoid RBC transfusion due to increased viscosity

TUMOR LYSIS SYNDROME

- **rapid release of intracellular metabolites from necrotic tumor cells in quantities that exceed the excretory capacity of the kidneys.**
  - Increased serum K, P, uric acid
- **Cardiac arrhythmias, renal failure, seizures, coma, disseminated intravascular coagulation (DIC) and death**



TLS: DIAGNOSIS

- **Laboratory TLS: 2 or more of below levels**
  - Uric acid >8 mg/dl,
  - Potassium > 6 mg/dl,
  - Phosphate > 6.5 mg/dl
  - Calcium < 7 mg/dl
- **Clinical TLS: laboratory TLS + 1 of below**
  - renal insufficiency, cardiac arrhythmias, seizures, or sudden death

Table 30-1 Patient Stratification by Risk for Tumor Lysis Syndrome for Various Types of Cancer

	Risk		
Type of Cancer	**High**	**Intermediate**	**Low**
NHL	Burkitt, lymphoblastic, B-ALL	DLBCL	Indolent NHL
ALL	WBC ≥ 100,000/mm^3	WBC 50,000-100,000/mm^3	WBC ≤ 50,000/mm^3
AML	WBC ≥ 50,000/mm^3	WBC 10,000-50,000/mm^3	WBC ≤ 10,000/mm^3
CLL		WBC 10,000-100,000/mm^3 Tx w/fludarabine	WBC ≤ 10,000/mm^3
Other hematologic malignancies (including CML and multiple myeloma) and solid tumors		Rapid proliferation with expected rapid response to therapy	Remainder of patients

TLS: MONITORING & TX

- **Serum LDH, BUN/Cr, electrolyte, uric acid, VBG**
- **Urine pH & specific gravity**
- **EKG, Urine output and Body weight**
- **Prevention:**
  - Fluid and hydration: 2L/m^2/day
  - Urine output: 3ml/kg/hr should be maintained
  - Diuretics such as lasix
  - Allopurinol: 10mg/kg/day
  - Rasburicase: 0.15~0.2 mg/kg/dose QD
    (禁忌：G6PD deficiency)

TLS: TX FOR HYPERURICEMIA

- **Results in precipitation of uric acid in distal renal tubules causing a decrease in glomerular filtration rate and ultimately acute renal failure.**

- **IV hydration**
- **Alkalization**
- **Allopurinol**
- **Rasburicase**
- **Acetazolamide (Diamox) or lasix**

TLS: TX FOR HYPERKALEMIA

- **Remove K from IVF, diet or drug**
- **Mild: IVF, Lasix, Kayexelate**
- **Moderate to severe (>6):**
  - EKG monitoring
  - Insulin + dextrose
  - Sodium bicarbonate (jusomin)
  - Dialysis
  - Life threatening: Calcium gluconate

TLS: TX FOR HYPERPHOSPHATEMIA

- **Hydration and diuresis**
- **Aluminum hydroxide**
- **Calcium carbonate**
- **Sevelamer hydrochloride (phosphate binder)**

- **Avoid alkalization to prevent crystallization**
- **Dialysis**

TLS: TX FOR HYPOCALCEMIA

- **At a calcium X phosphate product of >60 renal calcification can occur and exacerbate renal damage.**
- **0.5– 1.0 ml/kg of 10% calcium gluconate should be given in case of tenany**

TLS: TX FOR RENAL FAILURE

- **Indication for dialysis**
  - Presence of hyperphosphatemia (>6 mg/dL) and hypercalcemia
  - An estimated glomerular filtration rate (GFR) less than 50%
  - Persistent hyperkalemia with QRS interval widening and/or level exceeding 6 mEq/L
  - Severe metabolic acidosis
  - Volume overload unresponsive to diuretic therapy
  - Anuria and overt uremic symptoms (i.e. encephalopathy)
  - Severe symptomatic hypocalcemia
  - Hypertension (BP .150/90) and inadequate urine output at 10 hours from start of treatment
  - Congestive heart failure.

HYPERCALCEMIA

- **Serum Ca > 12mg/dL**
- **Rare**
- **Etiology**
  - Osteolytic bone lesions (particularly in T-cell leukemia and lymphoma)
  - Bone demineralization secondary to parathyroid-like hormone (PTHrP) produced by tumors (paraneoplastic syndrome)
  - Defect in renal excretion.
- **症狀：噁心，嘔吐，便秘，頻尿，肌肉無力，心律不整，昏迷**

HYPERCALCEMIA: TX

- **Hydration and lasix, Stop Ca-containing medication**
- **Prednisone 1.5–2.0 mg/kg daily**
- **Bisphosphonates such as pamidronate (Aredia) 0.5–1 mg/kg IV over 4–6 hours**
- **Dialysis**

SIADH

- **Syndrome of Inappropriate Antidiuretic Hormone Secretion**
- **Hyponatremia, hyposmolality and water intoxication**
- **Risk: stress, pain, surgery, mechanical ventilation, infections, CNS and pulmonary lesions, lymphomas and leukemias.**
- **ARD of Medication: vincristine, vinblastine, cyclophosphamide, ifosfamide, cisplatin and melphalan; or overhydration**

SIADH

- **Features: Oliguria, Body weight gain; headache, nausea, fatigue; stupor, seizure and coma**
- **Lab finding:**
  - Serum osmo < 280, Na <130
  - urine osmo > 500, Sp Gr increased
- **Treatment**
  - Fluid restriction, diuretics
  - Insensible loss 500ml/m2/day
  - In severe case: 3% NaCl, but slowly

BY ANATOMIC REGION

SUPERIOR VENA CAVA SYNDROME

- **Compressed or thrombotic SVC**
- **Usually by anterior mediastinum mass**
  - Hodgkin lymphoma
  - Non-Hodgkin lymphoma
  - Teratoma
- **Sometimes trachea compression is also noted**
- **臉紅脖子粗，Cyanotic change of upper extremity**
- **Collateral vein engorged, suffusion of the conjunctiva**
- **Cough, wheezing or stridor, hoarseness, dyspnea, dysphagia, chest pain**

SVCS: MANAGEMENT

- **Prevent supine position, sedation**
- **CXR or Chest CT**
- **CBC/DC, LDH, uric acid, AFP/beta-HCG**
- **Assess anesthesia risk (such as for diagnostic procedure)**
- **Treatment:**
  - Steroid: prednisolone / methylprednisolone
  - Emergent radiation therapy
  - Chemotherapy
  - Anticoagulants or thrombolytic agents

ABDOMINAL EMERGENCY

- **Esophagitis: the most common gastrointestinal (GI) problem in oncology patients.**
- **Gastric hemorrhage: especially in patients on corticosteroid therapy.**
- **Typhlitis: seen only in neutropenic patients.**
- **Perirectal abscess: in prolonged neutropenia.**
- **Hemorrhagic pancreatitis: especially in patients on L-asparaginase therapy.**
- **GI obstruction: Burkitt's lymphoa of the ileum, pelvic sarcoma or presacral teratoma, Medication…**
- **Massive hepatic enlargement from tumor**

TYPHLITIS（"盲腸"炎）

- **Typhlitis（"盲腸"炎）:**
  - A necrotizing colitis localized in the cecum
  - Severe neutropenia
  - Right lower quadrant pain with local mass lesion
  - A result of bacterial or fungal invasion of the mucosa, then full-thickness infarction to perforation -> peritonitis -> septic shock.
  - Pseudomonas species, Escherichia coli, other Gram- negative bacteria, Staphylococcus aureus, α-hemolytic Streptococcus, Clostridium, Aspergillus, and Candida

TYPHLITIS

- **Typhlitis management:**
  - Image study: KUB, Abdominal echography, CT
  - NPO for bowel rest, NG for decompression
  - IVF hydration and electrolyte supplement
  - Broad-spectrum antibiotics (anaerobic and Gram-negative coverage)
  - Surgical intervention when:
    - Persistent GI bleeding
    - Perforation
    - Uncontrolled sepsis

PERIRECTAL ABSCESS

- **Perirectal abscess**
  - Receiving Chemo, R/T, with prolonged neutropenia
  - staphylococci, streptococci, E. coli, Pseudomonas and anaerobic
  - Anorectal pain, tenderness and painful bowel movements
  - An abscess or draining fistula may be present
  - Management:
    - Broad-spectrum antibiotics, G-csf, Sitz bath
    - Surgical intenvention

PANCREATITIS

- **Pancreatitis**
  - Asparaginase, glucocorticoid
  - RUQ pain, emesis
  - Amylase, lipase, CT or abdominal echography
  - NPO, IVF, antibiotics for necrotizing pancreatitis
  - Complication: Pseudocyst, abscess, insufficiency and chronic change

GI BLEEDING

- **Upper GI bleeding**
  - Stress ulcer, steroid use, IICP, R/T …
  - Hematemesis, Melena
  - H2-blocker or PPI, endoscopy
- **Lower GI bleeding**
  - Infection colitis, typhlitis, post-OP, hemorrhoid, perirectal infection, anal fissue
- **Correct Thrombocytopenia or coagulopathy**

NEUROLOGIC EMERGENCY SPINAL CORD COMPRESSION

- **Spinal cord compression**
  - 3~5% cancer patient
  - Sarcoma, neuroblastoma, germ cell tumor, lymphoma, leukemia and drop metastasis of CNS tumors
    - Paraspinal lesion with spinal invasion
  - Vertebral body collapse (compression fracture) due to medication

SPINAL CORD COMPRESSION

- **Clinical features**
  - Low back pain with tenderness in most pt
  - Lower limb weakness
  - Incontinence
- **Evaluation**
  - Spine X-ray, CT, MRI
- **Management**
  - Steroid to decrease local edema
    - Dexamethasone 1~2 mg/kg/day loading
    - 0.25~0.5 mg/kg/dose Q6H
  - Surgical: Laminectomy
  - Known diagnosis: Chemo, emergent R/T

IICP 腦壓增高

- **CNS tumors, blocked shunt, pseudotumor cerebri, infection.**
- **Vomiting, headache, unconsciousness, seizure**
- **Late: Cushing's triad: bradycardia, hypertension, irregular respiratory pattern**
- **Management:**
  - Head elevation
  - Fluid restriction: 75% maintenance
  - Steroid: Dexamethasone同上
  - Diuresis: Mannitol
  - Endotracheal intubation: hyperventilation
  - Consult NS: EVD, VP shunt revision, tumor resection

SEIZURE

- **CNS tumors, CNS leukemia, intrathecal chemotherapy, metabolic derangements**
- **Antiepileptic drugs:**
  - Diazepam (Valium): IV 0.25mg/kg , PR 0.5mg/kg
  - Lorazepam (Ativan): 0.05~0.1 mg/kg IV
  - Phenytoin (Dilantin): 10~20mg/kg bolus IVF,
  - Phenobarbital (Luminal): 10~20mg/kg bolus IVF